University of Cape Town · Department of Information Systems

INF4027W Information Systems Honours: Systems Development Project 2026

FreightProof SA

Verified chain of custody for South African road freight

Iteration 4 Documentation

Team 05

Ciaran Formby (Product Owner) · **[ADD STUDENT NUMBER]**

Tim Gultig (Scrum Master) · **[ADD STUDENT NUMBER]**

Thomas Davis (Developer) · **[ADD STUDENT NUMBER]**

Chiko Kasongo (Developer) · **[ADD STUDENT NUMBER]**

Industry partner: Bruce van Wyk, Load Factor Group

Supervisors: Elsje Scott, Ammar Canani, Maureen Tanner, Ridewaan Hanslo

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Glossary

Acronyms are also written out in full the first time they appear in the text.

Table 1: Glossary of terms and acronyms

Term	Meaning in this document
Anchoring	Publishing the SHA-256 fingerprint of a record to the Hedera Consensus Service so that any later change to the record can be detected. Only the fingerprint leaves FreightProof, never the record.
API (Application Programming Interface)	The defined way one system requests data from another.
Arrival	The phase, new in Iteration 4, in which the driver records the seal as found at a stop before anything is opened.
Audit pack	A frozen, anchored evidence bundle for one trip or consignment, issued to an insurer or principal as a PDF and a verifiable web page.
BO, PO, SO	Business objective, project objective and system objective (§3).
BP	Business problem (§2).
CC	Custody Chain Capture: the driver- and receiver-side use cases.
CI (Continuous Integration)	Automated checks that build and test every change before it joins the shared code.
Client portal	The public web application where an audit-pack recipient views and independently verifies a pack.
Consignment	One waybill on a trip and the parcels it lists.
Corroboration	Comparing independent position sources (driver phone, horse tracker, trailer trackers, precinct) to support or question a recorded action.
"Could not check"	The recorded result when a comparison cannot be made, for example because a position is missing or stale. It is never treated as a match or a mismatch.
Dispatcher / Admin-Dispatcher	The operations users who create and monitor trips and review exceptions. Only an Admin-Dispatcher may change fleet and precinct records.
ECTA	Electronic Communications and Transactions Act 25 of 2002.
Exception	A recorded problem raised by a driver, a dispatcher or the system, with a severity of info, warning or critical.
Fingerprint	A SHA-256 (Secure Hash Algorithm, 256-bit) hash of a record. It is not a digital signature (§7.2).
Geofence	A circular boundary around a precinct used to check whether a position is inside it.
HCS	Hedera Consensus Service, the public ledger service that timestamps fingerprints.
Horse	The truck tractor unit that pulls one or more trailers.
IDV (Identity Verification)	A document-and-face check of a person, provided by Didit and used only for consignments that require it.
LFG	Load Factor Group, FreightProof's industry partner.
Manifest	The client's Parcel Perfect list of waybills for one departure. FreightProof creates a loaded trip from it.
MVP (Minimum Viable Product)	The smallest complete system that runs a trip end to end, with its limitations stated.
Organisation	One tenant of FreightProof: an operator, a principal or both. Users and data are isolated per organisation (§1.3).

Term	Meaning in this document
OTP (One-Time Password)	A short code sent to a phone or e-mail address to confirm a sign-in or an access request.
Override	A dispatcher action that closes a phase the driver cannot complete. A reason is mandatory and the phase is shown as overridden, never as verified.
Phase	One step in a trip's custody sequence: trip creation, activation, loading, departure, in transit, arrival, unloading or confirmation.
POD (Proof of Delivery)	The evidence captured when cargo is handed over at the destination.
POPIA	Protection of Personal Information Act 4 of 2013.
Precinct	A depot, warehouse or client site where handovers take place, with a location and geofence.
Principal	The organisation, for example a courier, that owns the consignments and contracts the transporter.
PWA (Progressive Web App)	A web application that installs on a phone and keeps working with poor connectivity.
QR code (Quick Response code)	A scannable square barcode. The driver's phone shows one that the receiver scans to confirm delivery.
Receiver	The person at the destination who confirms delivery on their own phone.
TLM	Trip Lifecycle Management: the dispatcher-side use cases.
Waybill	The document identifying a consignment and its parcels.

1. Project Background

1.1 Industry Setting

South African road freight moves goods between depots, warehouses and customers over long corridors such as Johannesburg to Durban. A single movement involves several organisations. The principal, such as a courier, owns the consignment. The transporter supplies the truck and driver. The precinct operator controls the gates, and the receiver takes delivery. When something goes wrong, the insurer carries the consequence. Each time cargo passes from one party to another there is a handover, and responsibility for the cargo moves with it.

Cargo theft, short deliveries and disputed deliveries are a recognised operational risk. The South African Police Service (SAPS) recorded 1,589 truck hijackings during 2024/25, about 4.4 a day (South African Police Service [SAPS], 2025). When an incident occurs, the parties must reconstruct events, often weeks or months later, from records each of them keeps separately. Table 2 lists the systems operators already use and how FreightProof connects to each one at the end of Iteration 4.

Table 2: Existing systems and providers, and their final integration mode

System	What it records	Role for FreightProof	Iteration 4 mode
Pulsit (fleet telematics)	Positions of the horse and trailers from fitted trackers	A position source the driver cannot influence	Integration code complete; simulated feed, no partner credentials
Parcel Perfect (parcel management)	Manifests, waybills and parcel barcodes: what should be on the truck	Supplies the expected cargo when a trip is created from a manifest	Waybill interface built; manifest feed simulated
Warehouse scanning	Which parcels were physically scanned at a depot	The observed cargo, compared with the expected cargo	Simulated scan feed
Didit (identity verification)	Identity document and face checks	Optional receiver check on consignments that require it	Live in the deployed system; off by default
Hedera Consensus Service	Ordered, timestamped messages on a public ledger	Independent timestamp for record fingerprints	Live on the Hedera test network

Each system performs its own function well. The gap lies between them at the moment of handover. Each keeps a partial record keyed to its own reference: Parcel Perfect to the waybill, Pulsit to the vehicle and the gate to the entry event. No shared record ties them to one trip, and no single party owns the whole journey or all of its records.

1.2 Industry Partner and Sponsorship

FreightProof is a self-sponsored Honours project. It is designed as a product venture: an evidence service that road-freight operators would subscribe to. Its industry partner is Load Factor Group (LFG), a Gauteng-based operator that supplies vehicles, drivers and line-haul transport to principals on depot-to-depot routes. LFG uses Pulsit for fleet tracking and Parcel Perfect for waybills, and carries cargo that principals have already packed and manifested.

LFG's director, Bruce van Wyk, is the project's domain expert and LFG is the intended first customer. LFG does not commission, fund or own the system; the team sets the scope and is responsible for the design. Iteration 4 also drew on seven research interviews held in September 2026 with a freight operator, a logistics software provider, a cargo insurer, a legal and privacy specialist, a digital-forensics practitioner and two retail logistics groups. Their findings shaped the audit pack and the exception workflow (§6).

1.3 System Overview

FreightProof records evidence of each handover in a road-freight trip and makes it available, intact, to the people who later need it. It does not replace any system already in use. It sits between them and captures what none of them owns. It records what happened; it does not dispatch armed response, reroute drivers or schedule vehicles.

Trip and phases. A dispatcher creates a loaded trip by entering the client's Parcel Perfect manifest number and choosing the driver, horse and trailers. FreightProof imports every waybill and parcel on that manifest and generates a phase plan from the trip's stops. The driver completes each phase in the Driver PWA: activation, loading, departure, in transit, arrival, unloading and confirmation. At each stop phase FreightProof records the driver's phone position and the horse and trailer tracker positions, and compares them with each other and with the precinct. It also checks parcels and seals where the phase requires it. Disagreements become exceptions for review; they never stop the driver.

Delivery. At the final stop the receiver scans a rotating QR code shown on the driver's phone and confirms delivery on their own phone, with a typed name, a drawn signature and their position. A document-and-face check is requested only when the consignment is flagged as high value or dangerous.

Integrity. Every completed phase, every critical exception, every exception review, every cancellation and every issued audit pack is anchored. FreightProof calculates a SHA-256 fingerprint of the record and publishes only that fingerprint to the Hedera Consensus Service (HCS). A later change to the record no longer matches its anchored fingerprint. Identities, positions and photographs stay in FreightProof's database and file storage in the Supabase af-south-1 (Cape Town) region, in line with the Protection of Personal Information Act (POPIA).

Sharing evidence. A dispatcher can issue an audit pack: a frozen, anchored PDF and web page that an insurer or principal opens through a share link and an e-mailed one-time code. The recipient's own browser re-checks the pack against Hedera, so the recipient does not have to trust FreightProof's servers.

Organisations. FreightProof is multi-tenant. An *organisation* is one company using the system: an operator (such as LFG), a principal (such as a courier client) or both. Every user belongs to one organisation, and every read and write is limited to that organisation's records; another organisation's record reads as "not found". The one deliberate exception is precincts. A principal's depot can be shared with the operators that deliver to it, who may read it but not change it. Receivers and audit-pack recipients are not members of any organisation: they act through single-use links and never see other records.

2. Business Problems

Six business problems (BP) were identified with LFG in Iteration 2 and refined after each review. Table 3 summarises them; the paragraphs that follow give the detail. How each problem is addressed is traced in §3.4, after the objectives are defined.

Table 3: Business problems in summary

ID	Problem in one line	Who bears the loss
BP-1	Evidence of each handover is split across parties and systems	Transporter and insurer, through slow and contestable investigations
BP-2	Nobody can show reliably who performed or accepted a handover	Transporter, who cannot rebut a disputed delivery
BP-3	Unplanned driver, vehicle or trailer substitution looks like a planned one	Transporter, who carries the theft or dispute
BP-4	Expected and received quantities are compared on paper, if at all	Transporter, who usually absorbs short deliveries
BP-5	Events must be reconstructed weeks or months later from incomplete records	Insurer and transporter, through claims settled on partial evidence
BP-6	Unresolved theft and disputes raise costs and weaken commercial terms	Transporter, through higher premiums and lost contracts

BP-1 Fragmented evidence at handovers. Dispatchers, principals and insurers each hold part of the record of a handover. An investigator must find and connect records from transport systems, warehouse systems, trackers, cameras and documents. The research interviews confirmed that this assembly, not the absence of any single record, is what makes investigations slow.

BP-2 Uncertain identity at handovers. A driver is checked at registration, but nothing reliably shows that the same person was present at each later handover. Receivers are often identified only by a name on paper. The transporter therefore cannot confidently rebut or pursue a disputed handover.

BP-3 Undetected substitution. Planned driver changes and trailer swaps occur on long corridors, for example at the Harrismith exchange point on the Johannesburg–Durban route. Without a timestamped record, an unplanned substitution looks like a planned one.

BP-4 Quantity disputes at delivery. The expected quantity sits in Parcel Perfect and the received quantity on a paper proof of delivery. Without a neutral comparison, the transporter often absorbs the loss or the cost of the dispute.

BP-5 Late reconstruction. Insurers and investigators may ask for evidence long after a trip, when documents are lost, camera footage has expired and tracker history is no longer detailed enough.

BP-6 Business and insurance exposure. Unresolved theft and disputes cost money directly and weaken the operator's position with principals and insurers. Better insurance terms are a long-term outcome only, and FreightProof does not claim to deliver them.

3. Business, Project and System Objectives

3.1 Business Objectives

The business objectives (BO) belong to FreightProof as a venture. FreightProof succeeds as a business only if operators such as LFG adopt it, so each objective is stated as FreightProof's goal and measured by an outcome for its customers. Following Iteration 3 feedback, system capabilities have been removed from the business objectives; they appear only in §3.3.

BO-0 (parent): establish FreightProof as a sustainable evidence service for South African road-freight operators, with LFG as the reference customer.

Table 4: Business objectives and how each will be measured

ID	Business objective	Measure and baseline
BO-1	Secure a reference customer by completing a pilot with LFG on the Johannesburg–Durban lane	A completed pilot with LFG's written acceptance of the evidence it produced. The project ends before a pilot can run, so this is the first post-project milestone.
BO-2	Reduce the time customers spend assembling evidence for a dispute or claim	Hours from request to complete evidence pack. LFG's current figure will be measured at the start of the pilot; no figure is assumed.
BO-3	Reduce the losses customers absorb from short deliveries, substitution and disputed handovers	Share of disputes resolved with FreightProof evidence, and value written off per quarter, compared with the operator's records for the previous year
BO-4	Make customers' evidence acceptable to third parties	Audit packs accepted by an insurer or principal without further evidence requests; recipients able to verify a pack without FreightProof's help
BO-5	Run the service at a cost per trip that an operator will pay	Hosting, ledger and identity-check cost per trip, from provider invoices, against a price agreed with the pilot customer

3.2 Project Objectives

The project objectives (PO) describe what this Honours project set out to deliver in support of the business objectives. Table 5 gives their status at the end of Iteration 4.

Table 5: Project objectives and final status

ID	Project objective	Status	Evidence
PO-1	Capture handover evidence: record, at each phase, which registered account acted, where, when and with what supporting evidence	Achieved	Phase ledger, location corroboration and photo evidence (§6.5, §8)
PO-2	Preserve evidence and detect changes, so that a later change to an anchored record can be detected	Achieved	Every phase, critical exception, review, cancellation and audit pack anchored (§7.2)
PO-3	Detect discrepancies and substitution indicators by comparing independent sources. The aim is detection, not prevention	Achieved, with simulated feeds	Tracker, precinct, parcel and seal comparisons (§6.5); Pulsit and scan feeds simulated

ID	Project objective	Status	Evidence
PO-4	Deliver a working MVP with stated limitations	Achieved	Deployed system (§7.1) and known limitations (§5.4)

3.3 System Objectives

Table 6 states the system objectives (SO), their acceptance criteria and their status at the end of Iteration 4. The configured values, such as distances and time limits, are explained in §7.3.

Table 6: System objectives, acceptance criteria and status

ID	System objective and acceptance criterion	Iteration 4 change	Status
SO-1	**Evidence anchoring.** Every completed phase, critical exception, exception review, trip cancellation, phase override, audit pack and fleet or precinct change is fingerprinted and anchored, with a receipt targeted within 60 s. A ledger outage never stops a driver; failed submissions are retried.	Every phase anchored (FP-270); exceptions, reviews and cancellations anchored (FP-290)	Delivered and tested; **[ADD: share of receipts within 60 s from the load test]**
SO-2	**Location corroboration.** At each stop phase the phone, horse tracker and every trailer tracker are compared with the precinct geofence, and the phone with the horse. A missing or stale position is recorded as "could not check", never as a mismatch.	Trailer trackers and on-road tracker check added (FP-270, FP-272)	Delivered with a simulated Pulsit feed
SO-3	**Manifest and parcel traceability.** A loaded trip is created from a Parcel Perfect manifest; expected and scanned parcels are compared at loading and unloading; any parcel can be traced by barcode.	Manifest-first creation (FP-281); parcel search (FP-149)	Delivered with simulated Parcel Perfect and scan feeds
SO-4	**Receiver acknowledgement.** Delivery is confirmed on the receiver's own phone through a single-use, rotating QR link, with a typed name, signature and position. An identity check is requested only for consignments that require it, and never prevents confirmation. The driver cannot close a delivery without the receiver's confirmation.	Identity check opt-in per consignment (FP-293); receiver confirmation enforced (FP-273); selfie stored (FP-316)	Delivered
SO-5	**Exception recording and review.** Every exception, of every severity, is reviewed by a dispatcher, who claims it, records an outcome and contact method, and cannot alter the original report.	All exceptions reviewed; claim, release and batch review (FP-280)	Delivered
SO-6	**Seal integrity.** The seal number and photograph are recorded at departure and inspected at arrival, before the doors open. A mismatch, damaged seal or missing seal raises a critical exception.	Arrival phase (FP-270); seal number read from the photo (FP-300)	Delivered
SO-7	**Operational analytics.** Activity, timeliness, problems, evidence, routes, sites and cost are calculated from current records, and every figure states its source.	Cost analytics (FP-294); today's incident map and sources (FP-296); AI brief (FP-295)	Delivered
SO-8	**Evidence sharing.** A dispatcher issues an audit pack that a named recipient can open with a share link and e-mailed code, and verify independently against the ledger. Access is logged and revocable.	New (FP-279, FP-291, FP-292)	Delivered

3.4 Traceability from Problems to Objectives

Table 7 traces each business problem to the business objectives it threatens, the system objectives that address it and the use cases in §6.5 where the evidence is captured.

Table 7: Traceability from business problems to objectives and use cases

Problem	BO	SO	Use cases	Remaining limitation
BP-1 Fragmented evidence	BO-2, BO-4	SO-1, SO-2, SO-8	TLM 1.12, TLM 1.5, CC 1.4	Gate-security and camera records are not integrated
BP-2 Uncertain identity	BO-3, BO-4	SO-4	CC 1.10, CC 1.8	Driver identity rests on phone sign-in; the receiver check is optional by design
BP-3 Undetected substitution	BO-3	SO-2, SO-5	CC 1.11, TLM 1.9, TLM 1.14	Tracker positions are simulated; location supports an account but does not prove identity
BP-4 Quantity disputes	BO-2, BO-3	SO-3, SO-6	TLM 1.2, TLM 1.13, CC 1.11	Scans come from a simulated feed; pallet-level counts are not modelled
BP-5 Late reconstruction	BO-2, BO-4	SO-1, SO-8	TLM 1.12, TLM 1.5	Evidence held outside FreightProof, such as camera footage, is referenced but not preserved
BP-6 Business exposure	BO-1, BO-5	SO-7	TLM 1.10	No pilot data; reductions in loss or premiums are not measured or claimed

4. Risk Assessment

Risks are scored as in Iteration 3: likelihood (L) and impact (I) each run from 1 to 5, and the score is L × I. Critical is 15–25, High 8–14, Medium 4–7 and Low 1–3 (Team05, 2026b, §5.1, p. 43). Each risk has one owner and one treatment, summarised as what reduces it (mitigate), the signal that it is happening (monitor) and the response (manage).

Following Iteration 3 feedback, the register is now a single table on one landscape page (Table 9), and items that were really unfinished features have been removed. They are listed as known limitations in §5.4. Five project risks have been added (R-19 to R-23). Every status uses the legend below.

Table 8: Risk status legend

Status	Meaning
New	First recorded in Iteration 4
Open	Unchanged since Iteration 3: same wording and score
Changed	The score has changed since Iteration 3; the arrow in the Δ column shows the direction
Realised	The risk has happened; the treatment describes how it is being handled
Closed	No longer applies; the reason is given below the register

Closed since Iteration 3. R-08 (guard resistance) remains closed: guards have no accounts. R-11 (delivery closed without a receiver confirmation) is closed because the server now refuses a confirmation phase that has no receiver handover (FP-273). R-17 (late or stalled trips not detected) was an unfinished feature rather than a risk: the live trip list now shows each trip's last update (FP-289), and the remaining gap, an automatic lateness alert, is listed in §5.4.

Table 9: Risk register, Iteration 4 (Δ = change in score since Iteration 3)

ID	Risk (cause → event → consequence)	Category	L×I	Δ	Status	Treatment: mitigate · monitor · manage	Owner
R-01	No independent scan source → counts rely on a simulated feed → loading unproven	Integration	4×4=16	=	Open	Replaceable scan interface; seal and receiver evidence · phase waits for scans · override with reason	Tim Gultig
R-02	Anchoring worker stops → receipts stay pending → no independent timestamp	Technical	3×4=12	↓	Changed	Retries and 5-minute recovery sweep · "receipt still owed" shown to all users · restart worker	Thomas Davis
R-03	Positions inaccurate, stale or simulated → wrong verdicts → false alarms or missed substitution	Technical	4×3=12	=	Open	Phone, horse and trailer compared; age limits; "could not check" · review inbox · phone driver, record outcome	Thomas Davis
R-04	Evidence altered before fingerprinting or while unanchored → change undetectable	Security	1×4=4	↓	Changed	Every phase, critical exception, review and pack anchored · failed verification · investigate as tampering	Chiko Kasongo
R-05	Offline queue loses evidence (blocked item, app data cleared) → incomplete record	Technical	2×4=8	↓	Changed	Blocked items kept (FP-276) · **automatic dispatcher alert** on undeliverable items · recapture or override	Ciaran Formby
R-06	Personal field added to an anchored payload → personal data on a public ledger	Legal	1×5=5	=	Open	Typed payload records; tests forbid personal fields in CI · failing test · block release	Ciaran Formby
R-07	Hedera outage → new trips and fleet changes refused → delay	External	2×4=8	=	Open	Phases complete and anchor later · "trip not created" error · retry after outage	Tim Gultig
R-09	Driver data processed beyond lawful scope → POPIA complaint	Legal	3×4=12	=	Open	Location only during trips; no driver scores · manual review · erasure and consent before real data	Chiko Kasongo
R-10	Driver app fails on untested Android devices → silent evidence gaps	Operational	3×3=9	=	Open	Company-issued devices; failures visible · incomplete phases on live list · replace device; override	Chiko Kasongo
R-12	Receiver identity check inaccurate or disproportionate → misleading evidence or POPIA complaint	Legal	1×4=4	↓	Changed	Check opt-in per consignment (FP-293); consent recorded · mismatch exceptions · follow up with receiver's company	Ciaran Formby
R-13	QR link forwarded or opened by someone else → remote handover	Security	3×4=12	=	Open	Single-use link, 20 s rotation, browser bound, position recorded · distant confirmations · treat as disputed	Ciaran Formby
R-14	Platform security weakness exploited → data exposed or altered	Security	2×5=10	↓	Changed	Database lockdown, per-user rate limits, upload limits (FP-271, 274, 308) · security review · fix before release	Ciaran Formby
R-15	Analytics misread or slow as data grows → wrong conclusions	Operational	2×3=6	↓	Changed	Source stated on every figure (FP-296); small samples flagged · load time · pre-compute if slow	Thomas Davis
R-16	Build or demonstration environment fails at the final presentation	Team	3×4=12	↓	Changed	Full suite in CI; deployment runbook; freeze from 19 Oct · red CI or failed health probe · local fallback	Thomas Davis
R-18	Driver account used by someone else (shared phone, SIM swap) → evidence misattributed	Security	3×4=12	=	Open	OTP sign-in; one session per driver; idle time-out · reported misuse · deactivate; treat as disputed	Tim Gultig
R-19	Work concentrated in two members → key-person dependency	Team	3×4=12	–	New	Per-person point limits; paired reviews · points per person · rebalance at planning	Tim Gultig
R-20	Carry-over and new scope crowd out hardening and documentation	Schedule	4×3=12	–	New	Carry-over reported separately; pointed documentation tickets; freeze · completion below 70% · cut lowest priority	Ciaran Formby
R-21	Members cannot explain AI-assisted code at code review → marks lost, defects missed	Team	3×4=12	–	New	Reviewed pull requests; author tests; module walk-throughs · unanswered review questions · pair on module	Tim Gultig
R-22	Partner credentials never arrive → integrations simulated at the final demonstration	External	5×3=15	–	Realised	Realistic mock feeds; one switch per integration · partner status · state the limitation; show the switch	Ciaran Formby
R-23	Large late merges break shared code → regressions near the deadline	Technical	3×3=9	–	New	CI checks migration head, layering and full suite · failed CI on dev · revert or fix before further merges	Ciaran Formby

5. Updated Project Plan and Revisions

This section is kept short, as the guidelines ask. Revisions made in Iteration 4 are marked **Revised** in the tables. The architecture revisions are described in §7 and the revised risks in §4.

5.1 Project Timeline

Table 10 shows every sprint in the project, taken from the Jira board. The gaps between sprints are the iteration reviews and the work that followed them: feedback analysis, research and backlog refinement.

Table 10: Project timeline by iteration and sprint

Iteration	Sprint	Dates (2026)	Focus
1	1–2	19 Apr – 17 May	Technical foundation: data model, ledger anchoring, sign-in, first trip flow
2	3–5	22 Jun – 10 Aug	Plan-driven phase lifecycle, Parcel Perfect interface, seal comparison, scan-driven loading
3	6–7	26 Aug – 17 Sep	Location corroboration, receiver handover, exception review, precincts, analytics
4	8 (**Revised** start)	29 Sep – 12 Oct	Hardening after the Iteration 3 review, Arrival phase, manifest-first trips, exception inbox, audit packs
4	9	12 Oct – 23 Oct	Live operations view, critical-evidence anchors, cost analytics, receiver model, final submission
—	Review	23 Oct; 26–28 Oct	Documentation submitted 23 Oct; presentation 26–27 Oct; code review 28 Oct

Revisions to the plan. Sprint 8 started on 29 September rather than straight after the Iteration 3 presentation, because the team first analysed the panel's feedback, held seven research interviews and verified the claims made on stage against the code. Sprint 9 ends on submission day, 23 October, and includes a structural code freeze from 19 October. Sprint lengths therefore vary; §10.4 shows how velocity is normalised for this.

5.2 Delivery Against Iteration 3 Commitments

Iteration 3 set out what Iteration 4 would deliver (Team05, 2026b, §2.4, p. 18). Table 11 reports each commitment.

Table 11: Iteration 3 commitments and their outcome

Commitment	Outcome	Evidence and remaining work
Live Pulsit and Parcel Perfect access, if possible	Not achieved	Partner introductions did not produce credentials in time. Both integrations remain simulated behind a configuration switch (R-22).
Live receiver identity-verification testing	Delivered, then re-scoped	Tested on the deployed system; after review feedback the check became opt-in per consignment (FP-293)
More detail on live trips	Delivered	Live trip list (FP-289), per-phase live timelines (FP-288) and photos appearing mid-phase (FP-287)

Commitment	Outcome	Evidence and remaining work
Parcel search by barcode (FP-149)	Delivered	Carried from Sprint 7 and completed in Sprint 8
Performance and resilience measurement	Delivered	Database timeouts (FP-275) and a load test (FP-299), reported in §9
POPIA controls	Partly delivered	Receiver biometrics off by default; database access locked down (FP-271). Driver data erasure is still a known limitation.
Johannesburg–Durban demonstration	Prepared	Demonstration trip and trip-following demo panel (FP-272); shown at the final presentation on 26–27 October

5.3 Iteration 4 Release Slice

The story-map backbone is unchanged: three user activities, *prepare and plan*, *execute custody* and *assure and prove*, with a *hand over to receiver* task column (Team05, 2026b, §2.2, Fig. 1, p. 17). Table 12 lists the Iteration 4 slice under each activity.

Table 12: Iteration 4 release slice by story-map activity

Activity	Iteration 4 stories
Prepare and plan	Create a trip from a manifest (FP-281); see vehicle and driver clashes before creating a trip (FP-139, FP-142); find precincts by address (FP-309)
Execute custody	Arrival phase and trailer geofence (FP-270); seal number read from the photo (FP-300); accept real seal formats (FP-278); breakdown affecting truck and trailer (FP-297); driver substitution (FP-83); offline evidence retained (FP-276)
Hand over to receiver	Receiver confirmation enforced (FP-273); identity check opt-in per consignment (FP-293); selfie stored (FP-316)
Assure and prove	Every exception reviewed (FP-280); anchored exceptions, reviews and cancellations (FP-290); live trip list and timelines (FP-287, FP-288, FP-289); audit packs and client portal (FP-279, FP-291, FP-292); parcel search (FP-149); cost analytics, today's incident map and AI brief (FP-294, FP-296, FP-295)

5.4 Known Limitations

These are stated limits of the final MVP, not risks. Several were listed as risks in Iteration 3.

- Pulsit tracker positions, Parcel Perfect manifests and warehouse scans come from simulated feeds. The integration code is complete and each feed is switched by one configuration setting.
- Geofences are circles; polygon precincts are not supported.
- There is no automatic lateness alert. Dispatchers see each trip's last update on the live trip list and act on it.
- Alerts reach dispatchers inside the application only. E-mail is used for audit-pack access codes, not for operational alerts.
- Driver data erasure and recorded driver consent are designed but not built.
- Exception follow-up notes and a separate evidence sign-off step were designed but not built; reviews record the outcome and the method of contact.

- Pallet or handling-unit counts are not modelled. Counts are at parcel and consignment level.
- Camera footage and gate-security records are not integrated.

6. Analysis and Design Work Products

This section presents the Iteration 4 versions of the package, class, use-case and state-machine diagrams, the business rules, the new and changed use-case narratives, and the database changes. Work products that did not change are referred to in the Iteration 3 document rather than reprinted.

6.1 Package Diagram

Figure 1 shows FreightProof's packages. Iteration 4 changed the structure in three ways. First, the backend's orchestration layer was split into one package per responsibility: trips, phases, exceptions, evidence, evidence anchors, handover, consignments, parcel trace, audit packs, analytics, fleet and verification. The former 1,962-line phase service is now a package of small modules. Second, a fourth front end was added: the client portal, where an audit-pack recipient views and verifies a pack. Third, the layering rule (request handlers → orchestration → integrations and ledger client → database) is now enforced automatically: an import-linter check in CI fails any change that skips a layer.

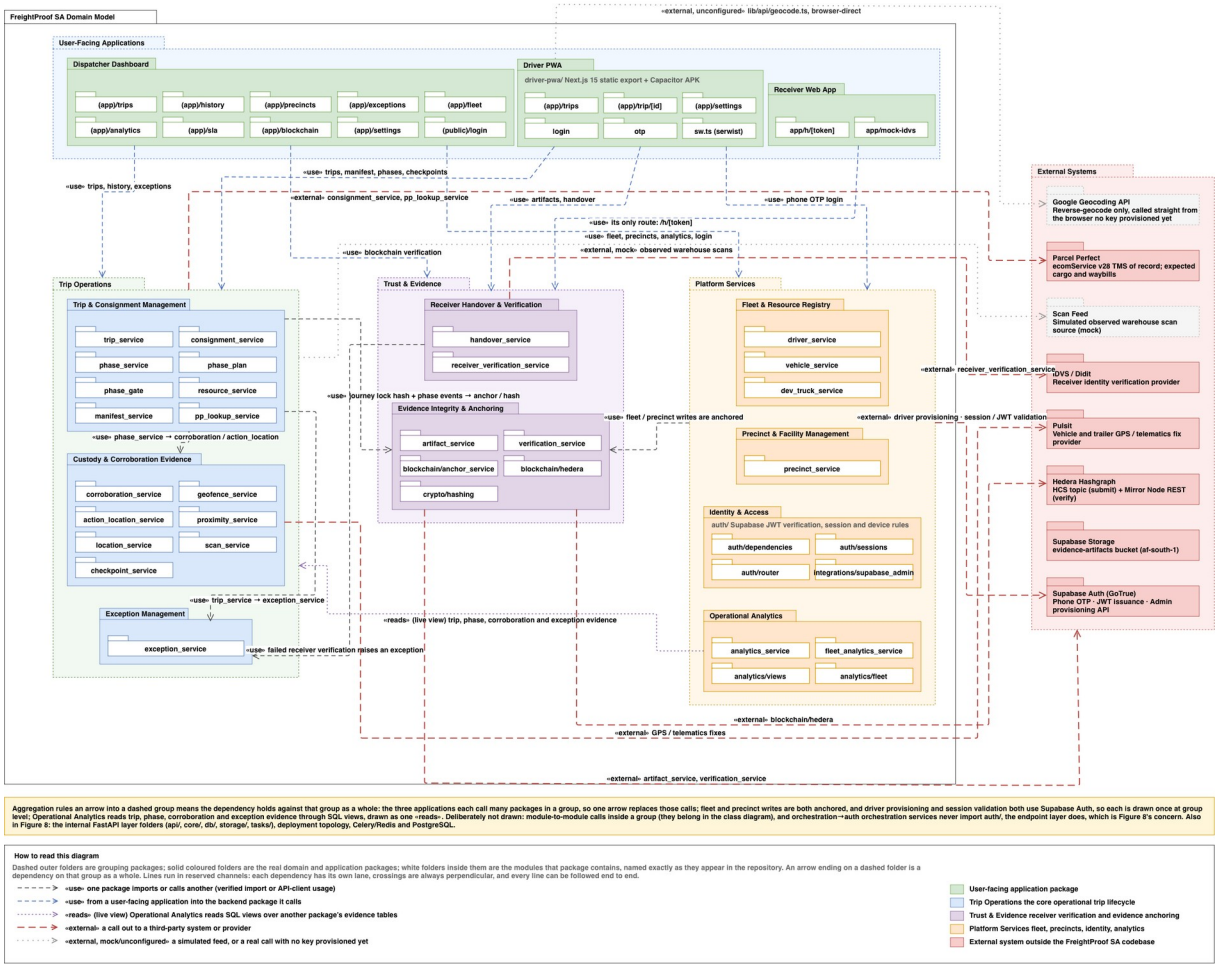


Figure 1: Package diagram

TEAM TO DO: This is the Iteration 3 diagram, kept as a stand-in. Replace it with the Iteration 4 version: show the orchestration packages listed above, the client portal as a fourth front end, audit packs, cost analytics and AI insights.

The dispatcher application, Driver PWA, receiver application and client portal depend on groups of packages rather than individual ones. External systems remain behind interfaces with simulated and

live implementations: Parcel Perfect supplies manifests and waybills, Pulsit supplies tracker positions, Didit performs optional receiver checks, SendGrid sends audit-pack access codes, OpenRouteService supplies planned routes, and Hedera receives fingerprints only.

6.2 Analysis Class Diagram

Figure 2 shows the analysis class model. It remains centred on Trip, with its ordered TripStops, its PhaseEvent ledger and its Consignments. Iteration 4 made five changes, which answer the Iteration 3 comment that parcel detail was modelled at too high a level:

- **Cargo hierarchy.** A loaded Trip now carries the key of the client's Parcel Perfect manifest. The hierarchy is *manifest → waybill (Consignment) → Parcel*. Each Consignment names its pickup stop and delivery stop, so a multi-stop trip can drop off and collect different waybills at each intermediate stop. The manifest snapshot taken at creation is stored on the trip-creation phase and covered by the journey lock.
- **Arrival.** PhaseEvent gains the seal condition found at arrival (intact, damaged or missing) and how the seal number was entered (read from the photo and confirmed, read and corrected, or typed).
- **Exception review.** TripException gains the claiming dispatcher and claim time, alongside the existing reviewer, outcome, contact method and note. An exception can link to several affected vehicles, so one breakdown can name the horse and its trailers.
- **Evidence sharing.** AuditPack, AuditPackAccessEvent, AuditPackEmailCode and IncidentDeclaration are new. EvidenceAnchorEvent records each anchored exception, review and cancellation with its own receipt.
- **Operations.** CostProfile holds an organisation's cost rates for cost analytics, and DriverSubstitution, present but unused in Iteration 3, now records driver handovers.

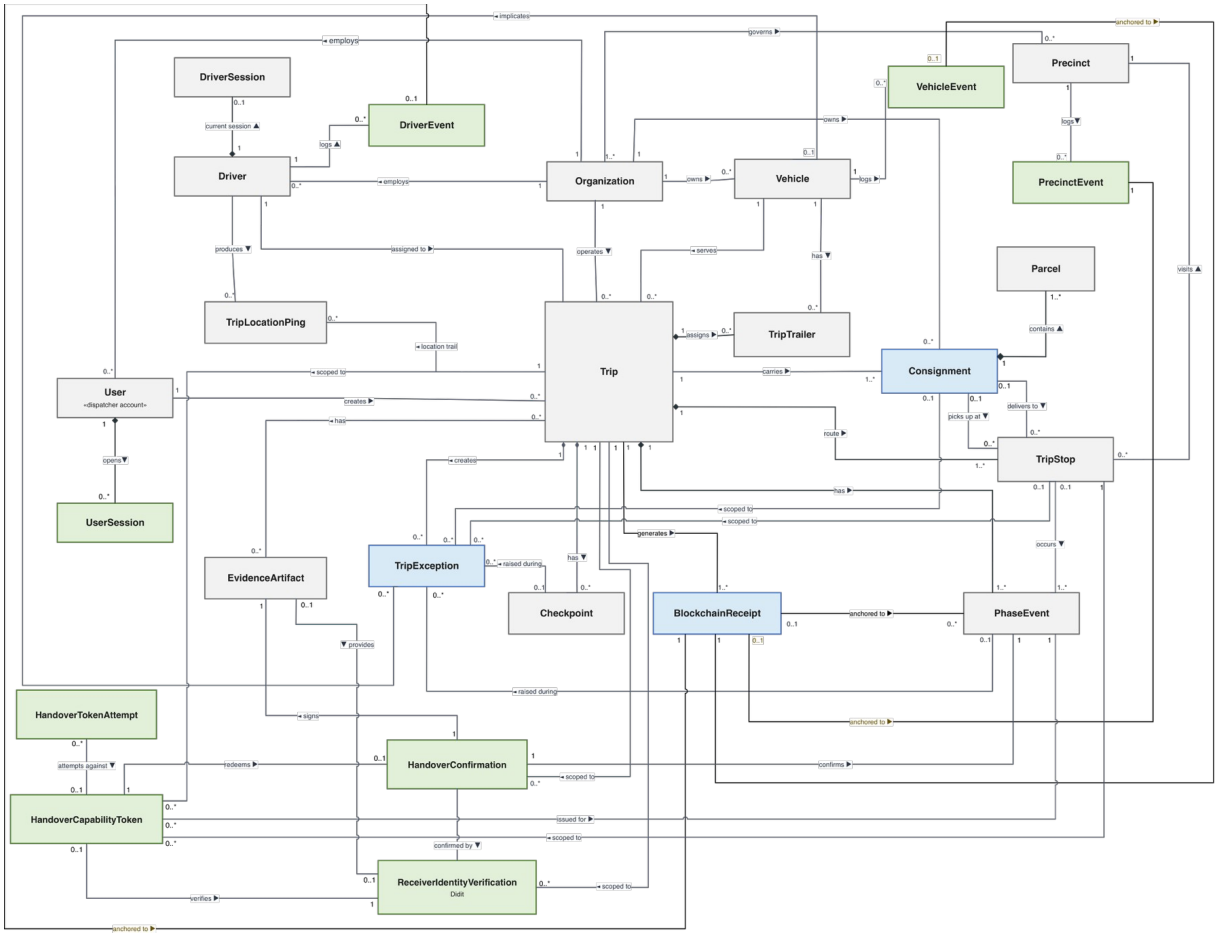


Figure 2: Analysis class diagram

TEAM TO DO: This is the Iteration 3 diagram, kept as a stand-in. Replace it with the Iteration 4 version: add the manifest → waybill → parcel hierarchy, the Arrival seal fields, exception claims, AuditPack, AuditPackAccessEvent, AuditPackEmailCode, IncidentDeclaration, EvidenceAnchorEvent, CostProfile and DriverSubstitution. Colour the new and changed classes.

Damage in transit is recorded as an exception against the trip, the affected consignment and the affected vehicles, with photographs. Pallet or handling-unit counts are not modelled: the partner could not confirm whether a handling unit sits between waybill and parcel, so counts stay at parcel level (§5.4).

6.3 Use-Case Model and System Boundary

Figure 3 and Figure 4 show the two use-case diagrams: Trip Lifecycle Management (TLM, dispatcher side) and Custody Chain Capture (CC, driver and receiver side).

System boundary. The boundary is one organisation's FreightProof account (§1.3). Dispatchers, Admin-Dispatchers and drivers are members of that organisation and act only on its records. Two actors are outside every organisation and have no account. The **Receiver** acts through a single-use link that exists only as a QR code on the driver's screen. The **Audit-pack recipient**, an insurer or principal, acts through a share link and an e-mailed code issued to them by name. Neither can list, search or open any other record. External systems (Parcel Perfect, Pulsit, the scan feed, Didit, SendGrid and Hedera) are shown in the architecture diagram (§7.1), not as actors.

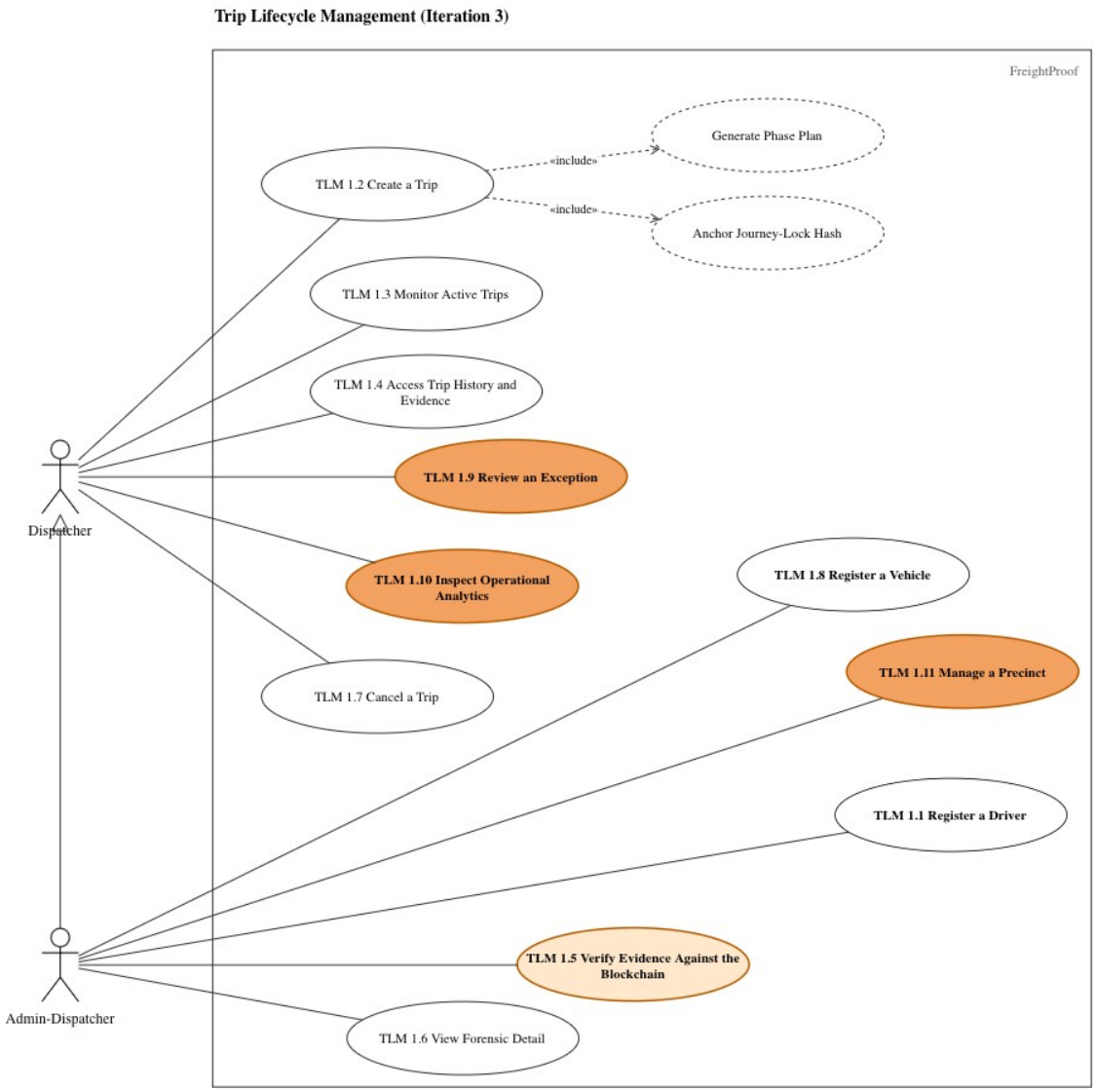


Figure 3: Trip Lifecycle Management use-case diagram



Figure 4: Custody Chain Capture use-case diagram

TEAM TO DO: Both use-case diagrams are the Iteration 3 versions. Add TLM 1.12, TLM 1.13, TLM 1.14 and CC 1.11, the Audit-pack recipient actor, and shade new and changed use cases as listed in the index table below.

Table 13 shows how each use case is treated in this document. Retained narratives are in the Iteration 2 document (Team05, 2026a) or the Iteration 3 document (Team05, 2026b, §3.4, pp. 27–35).

Table 13: Use-case retention and change index

ID	Use case	Treatment in Iteration 4
TLM 1.1, 1.8	Register a Driver; Register a Vehicle	Retained (Iteration 2)
TLM 1.2	Create a Trip from a Manifest	**Changed**: written in full (FP-281, FP-139, FP-142)
TLM 1.3	Monitor Active Trips	**Changed**: live trip list shows phase, last update, unreviewed exceptions and location check (FP-289); covered in §8.2
TLM 1.4, 1.6	Access Trip History; View Forensic Detail	Retained (Iteration 2)
TLM 1.5	Verify Evidence Against the Blockchain	Retained (Iteration 3, pp. 33–34); now also verifies exceptions, reviews and cancellations

ID	Use case	Treatment in Iteration 4
TLM 1.7	Cancel a Trip	**Changed**: cancellation is anchored (BR-8); otherwise retained
TLM 1.9	Review an Exception	**Changed**: written in full (FP-280, FP-290)
TLM 1.10	Inspect Operational Analytics	**Changed**: cost view, today's incident map and figure sources (FP-294, FP-296); covered in §8.5
TLM 1.11	Manage a Precinct	**Changed**: address search replaces typed coordinates (FP-309); otherwise retained (Iteration 3, pp. 29–30)
TLM 1.12	Issue an Audit Pack	**New**: written in full (FP-279, FP-291)
TLM 1.13	Trace a Parcel	**New**: written in full (FP-149)
TLM 1.14	Arrange a Driver Substitution	**New**: governed by BR-10 (FP-83)
CC 1.1–1.3, 1.5	Sign In; Activate Trip; Confirm Loading; Log Transit Checkpoint	Retained (Iteration 2)
CC 1.4	Seal and Depart	**Changed**: seal number read from the photo and confirmed by the driver (FP-300)
CC 1.6	Raise Exception or Panic	**Changed**: a breakdown can name several vehicles (FP-297); severity per BR-1
CC 1.7	Unload at Destination	**Changed**: the seal inspection moved to Arrival (CC 1.11)
CC 1.8	Confirm Delivery and Close Trip	**Changed**: replacement steps below (FP-273)
CC 1.9	Override a Blocked Phase	**Changed**: governed by BR-4
CC 1.10	Receive and Acknowledge Delivery	**Changed**: replacement steps below (FP-293, FP-316)
CC 1.11	Record Arrival	**New**: written in full (FP-270)

6.4 Business Rules

The Iteration 3 shared rules SR-1 to SR-9 remain in force (Team05, 2026b, Table 6, pp. 26–27). Table 14 adds the business rules that Iteration 3's review asked for: what happens when something goes wrong, who must be told, and what evidence is required. The "Enforced by" column separates what the system enforces from what the operator's procedure requires, so that no rule is claimed as automated when it is not.

Table 14: Business rules added in Iteration 4

ID	Rule	Enforced by
BR-1	**Severity.** An exception is critical if a person may be at risk, cargo may be lost or reached by someone outside the custody chain, goods went to the wrong party, or the trip cannot finish as planned. The system raises panic, a seal broken in transit, a seal mismatch, a damaged or missing seal at arrival and a trailer separated from its horse as critical; other exceptions start as warnings. When a reviewing dispatcher finds that a warning meets one of the criteria, for example cargo damage or a refused delivery, they handle it as critical under BR-3. Nobody may lower a severity.	System (severity by type); procedure (escalation)
BR-2	**Every exception is reviewed.** Every exception starts *unreviewed*. A dispatcher claims it, which records who and when, and may release it. The review records an outcome, the contact method and a note; the first review is final, and the original report is never changed. Critical exceptions are reviewed one at a time; warnings on one trip may be reviewed together. Critical exceptions and every review are anchored.	System (FP-280, FP-290)

ID	Rule	Enforced by
BR-3	**Who is told, and when.** A critical exception appears immediately in every dispatcher's live inbox in the organisation, so the driver does not need to phone it in. The reviewing dispatcher informs the operations manager at once for theft, hijacking, collision or injury. The operations manager notifies the principal within 24 hours of a security incident, in line with TAPA TSR (Transported Asset Protection Association, 2023), notifies the insurer within the period the policy requires, and opens a SAPS case for theft or hijacking. Each notification is recorded as an incident declaration with its time and reference number, and appears in the audit pack.	System (live inbox, incident declarations); procedure (notifications)
BR-4	**Overriding a phase.** Any dispatcher may override a phase the driver cannot complete, but must give a reason. The override is anchored as its own record and the phase is shown as overridden, never as verified. Overriding a departure, arrival or confirmation also requires an Admin-Dispatcher's approval, recorded in the reason, and any supporting document available (delivery note, photograph, SAPS case number) is attached to the trip.	System (reason, marking, anchoring); procedure (second approval, documents)
BR-5	**Disputed delivery.** If the scanned count differs from the manifest, a parcel-count exception is raised; a shortfall is handled as critical under BR-1. If the receiver refuses to sign or disputes the goods, the driver records a *delivery refused* exception with a reason and photographs. The reviewing dispatcher then decides whether the delivery is completed, overridden under BR-4 or cancelled under BR-8, and the principal is notified under BR-3. The refusal stays on the record, and a later inspection result is recorded as an incident declaration, a new, attributed record, never by editing the original.	System (exceptions, receiver confirmation); procedure (decision, notification)
BR-6	**Lost, stolen or dead phone.** The driver reports it to the dispatcher by any available means. Signing in on a replacement phone ends the old session automatically, and the dispatcher deactivates the account if the phone was stolen. Phases that cannot be completed are overridden under BR-4. Evidence captured between the loss and the deactivation is treated as disputed, and a theft is reported to SAPS.	System (single session); procedure (reporting)
BR-7	**Receiver identity.** By default the receiver confirms with a typed name, a drawn signature and their position, through a single-use QR link. A document-and-face check is requested only for consignments the dispatcher flags as high value or dangerous goods, and it never blocks delivery. Pre-registered authorised receivers nominated by the receiving organisation were evaluated and are recommended for the pilot, because identity would then be established once at onboarding rather than at the door.	System (FP-293)
BR-8	**Cancellation.** A reason is mandatory. The cancellation is anchored, the phases not reached stay pending as an honest record, and the manifest becomes available for a new trip.	System (FP-290)
BR-9	**Audit-pack access.** Only a dispatcher may issue an audit pack, to a named recipient, for a stated purpose. Its contents are frozen and anchored at issue. The share link expires, requires an e-mailed one-time code before first view, logs every view, and can be revoked.	System (FP-279, FP-291)
BR-10	**Driver substitution.** A substitution is a normal custody event, not an exception. A dispatcher arranges it; the outgoing and incoming drivers each record the handover, including the seal, on their own phones. The original driver's record is kept. A seal mismatch at the exchange is critical; an exchange outside the planned exchange point is a location warning.	System (FP-83)
BR-11	**Geofence size.** An Admin-Dispatcher sets each precinct's radius to cover the yard where trucks stand and handovers happen. The 200 m default plus 50 m tolerance is a starting value only.	Procedure

6.5 New and Changed Use-Case Narratives

Each narrative below states the actor, pre- and post-conditions, the main success scenario and the alternative flows. Alternative flows now cover the failure cases raised in the Iteration 3 review: a dead, lost or stolen phone; cleared app data; no signal; partner systems or the server being unavailable; and a receiver who cannot or will not confirm.

Table 15: TLM 1.2 – Create a Trip from a Manifest (changed)

Actor	Dispatcher	
Description	Create an evidence-bearing trip for one departure from the client's Parcel Perfect manifest, so that the cargo is imported rather than typed.	
Pre / Post	Pre: dispatcher signed in; the manifest exists in Parcel Perfect. Post: one trip with its stops, consignments, parcels and phase plan; the manifest snapshot and journey lock anchored.	
Main success scenario		
Step	**Actor action**	**System response**
1	Enters the manifest number.	Fetches the manifest and shows the client, route, planned times, waybills and parcel totals.
2	Chooses the driver, horse and trailers.	Shows an availability strip for the week and names any trip that already holds a chosen driver or vehicle (FP-142).
3	Reviews and confirms.	Checks again for clashes, creates the trip and phase plan, stores the manifest snapshot, and anchors the journey lock before saving.
Alternative and exception flows		
1a	Manifest not found, or its waybills belong to different clients: the system refuses and names the reason.	
1b	Parcel Perfect unavailable: the system refuses creation; no partial trip is saved.	
1c	A non-cancelled trip already exists for this manifest: the system refuses a second trip and names the existing one.	
2a	A precinct for the manifest's hub is not set up: the dispatcher chooses the precinct manually.	
3a	A clash appears between steps 2 and 3, for example another dispatcher assigned the horse: creation is refused with the clashing trip named (FP-139).	
3b	Hedera unavailable: creation is refused and nothing is saved (fail-closed), because the journey lock must be anchored.	

Table 16: TLM 1.9 – Review an Exception (changed)

Actor	Dispatcher	
Description	Assess every exception, record the conclusion and the contact made, and leave the original report unchanged.	
Pre / Post	Pre: dispatcher signed in; the exception belongs to the organisation. Post: one immutable review with reviewer, time, outcome and contact method; critical exceptions and the review anchored (BR-2).	
Main success scenario		
Step	**Actor action**	**System response**
1	Opens the unreviewed inbox, ordered by severity, or an exception from a trip timeline.	Shows type, severity, description, photographs, position, phase and stop.

2	Claims the exception.	Records who claimed it and when, and shows the claim to other dispatchers live.
3	Contacts the driver or site, then selects the outcome and contact method and writes a note.	Validates the review.
4	Submits.	Stores the review without changing the report, anchors it, and removes the exception from the inbox.
Alternative and exception flows		
2a	Another dispatcher has claimed it: the system shows who; this dispatcher may wait or ask them to release it.	
2b	The dispatcher cannot finish: they release the claim and the exception returns to the inbox.	
3a	Several warnings on one trip: the dispatcher reviews them together with one outcome and note; a review is stored for each. Critical exceptions cannot be batch-reviewed.	
4a	Another dispatcher submitted first: the system refuses the second review and shows the first.	
4b	Hedera unavailable: the review is saved, the receipt is marked owed and retried.	
4c	The incident requires notification: the dispatcher follows BR-3 and records an incident declaration.	

Table 17: TLM 1.12 – Issue an Audit Pack (new)

Actor	Dispatcher; Audit-pack recipient (insurer or principal, no account)	
Description	Give an outside party a frozen, verifiable evidence record of a trip or consignment without giving them access to FreightProof.	
Pre / Post	Pre: dispatcher signed in; the trip belongs to the organisation. Post: an anchored pack (PDF and web page); a share link issued to a named recipient; every access logged (BR-9).	
Main success scenario		
Step	**Actor action**	**System response**
1	Previews the trip's audit trail and chooses scope, purpose, recipient and whether to include the location trail.	Builds the pack from current records, marking each item as anchored or recorded only, and lists missing evidence.
2	Issues the pack.	Freezes the contents, renders the PDF, anchors the pack's fingerprint and shows the share link once.
3	Sends the link to the recipient.	No further action.
4	Recipient opens the link and enters the code sent to their e-mail address.	Logs the view and shows the evidence page: timeline, route map, photographs, exceptions and parties.
5	Recipient selects Verify.	The recipient's browser recomputes the fingerprints and checks them against the Hedera mirror service, then reports a match or a mismatch for each item.
Alternative and exception flows		
4a	Wrong code entered repeatedly: access is locked and the attempt logged.	
4b	Link revoked or expired: the page states this and shows no evidence.	
5a	Hedera mirror unavailable: the result is "could not check", never "verified".	

5b	A record changed after issue: the item shows as a mismatch, and records added after issue are counted separately.

Table 18: TLM 1.13 – Trace a Parcel (new)

Actor	Dispatcher	
Description	Find a parcel by its barcode and see where it has been, derived from the phase ledger.	
Pre / Post	Pre: dispatcher signed in. Post: none; the use case is read-only.	
Main success scenario		
Step	**Actor action**	**System response**
1	Enters or scans a parcel barcode.	Finds the parcel among the organisation's consignments.
2	Opens the result.	Shows the parcel's waybill, current state and a timeline of every phase and scan that involved it, with exceptions and evidence.
Alternative and exception flows		
1a	Barcode not found in this organisation: the system shows "not found" and does not reveal other organisations' parcels.	
2a	Parcel never scanned at a stop: the timeline shows the gap rather than inferring a location.	

Table 19: CC 1.11 – Record Arrival (new)

Actor	Driver	
Description	Record the seal as found at a stop before anything is opened, so that a break in transit is detected at the moment custody arrives.	
Pre / Post	Pre: the preceding in-transit phase is complete; the driver is at the Arrival phase. Post: Arrival completed once, with seal number, condition, photograph and positions stored and anchored.	
Main success scenario		
Step	**Actor action**	**System response**
1	Confirms arrival at the stop.	Records the phone, horse and trailer positions and compares each with the precinct geofence.
2	Photographs the seal.	Reads the seal number from the photograph and asks the driver to confirm or correct it (FP-300).
3	Confirms the number and records the seal as intact, damaged or missing.	Compares the number with the seal applied at departure.
4	Swipes to complete.	Stores the evidence, anchors the phase and opens the next phase (unloading, loading or confirmation).
Alternative and exception flows		
1a	Truck or a trailer outside the precinct: a location exception is raised for review; the driver continues.	
1b	No usable position, or no signal: recorded as "could not check"; the action is queued on the phone and sent when signal returns.	
3a	Number differs from departure, or seal damaged or missing: a critical exception is raised (BR-1) and the dispatcher is alerted at once.	

4a	Phone lost or dead before completion: handled under BR-6; the phase is overridden only under BR-4.
4b	App data cleared before the queued action was sent: the capture is lost on the phone, but nothing false is recorded. The phase stays pending on the server and on the live trip list, and the driver recaptures it.

Table 20: Changed narratives: replacement steps for CC 1.8 and CC 1.10

CC 1.8 – Confirm Delivery and Close Trip (Driver)	
Replaces step 5	The server completes the confirmation phase only if a receiver confirmation exists for that stop, and takes the signature from that confirmation. A signature uploaded by the driver is refused (FP-273).
Replaces alt. 3a	If the receiver has no suitable phone or connection, the driver cannot confirm; the dispatcher may override the confirmation only under BR-4, which requires an Admin-Dispatcher's approval and a signed paper delivery note photographed and attached.
New alt. 5a	If the receiver refuses to sign, the driver raises *delivery refused* under BR-5.
CC 1.10 – Receive and Acknowledge Delivery (Receiver)	
Replaces step 3	The system asks for an identity check only if the consignment is flagged as requiring one (BR-7). Otherwise the receiver goes straight to step 5.
Replaces step 4	When a check is required, the receiver consents and completes it; the selfie is stored securely before a "selfie only" result is recorded (FP-316). A scanned document number is not typed again.
Unchanged	Steps 1–2 and 5–8 and the alternative flows remain as in Iteration 3 (Team05, 2026b, pp. 30–31).

6.6 State-Machine Diagram

Figure 5 shows the life of one trip. Iteration 4 made the following changes, answering the Iteration 3 comments that a refused activation had no drawn path, that the "more stops planned?" decision had no false branch, and that the current stop could not load.

- **Created → Created on a refused activation.** When the driver asks to activate, the guards are checked: the trip is not being started early, the driver and the vehicles are not on another live trip, and no earlier trip of the driver's is due. If any guard fails, the trip stays in *Created* and the driver is shown the reason. This is drawn as a self-transition with the guard [activation refused].
- **One loop per stop, in a fixed order.** At each stop the order is: *Arrival* (every stop after the first), then *Unloading* [anything delivered here], then *Loading* [anything collected here]. A decision node then tests [more stops planned]: the [yes] branch goes to *Departure* and *In Transit*, and the [no] branch goes to *Confirmation*. The current stop therefore always loads before the vehicle departs.
- **Arrival** is a new state after every *In Transit*, where the seal is inspected (CC 1.11).
- **Cancelled** remains reachable from *Created* and *Active*, with a mandatory reason, and is now anchored (BR-8). **Closed** is reached only when every phase in the plan is completed or overridden.

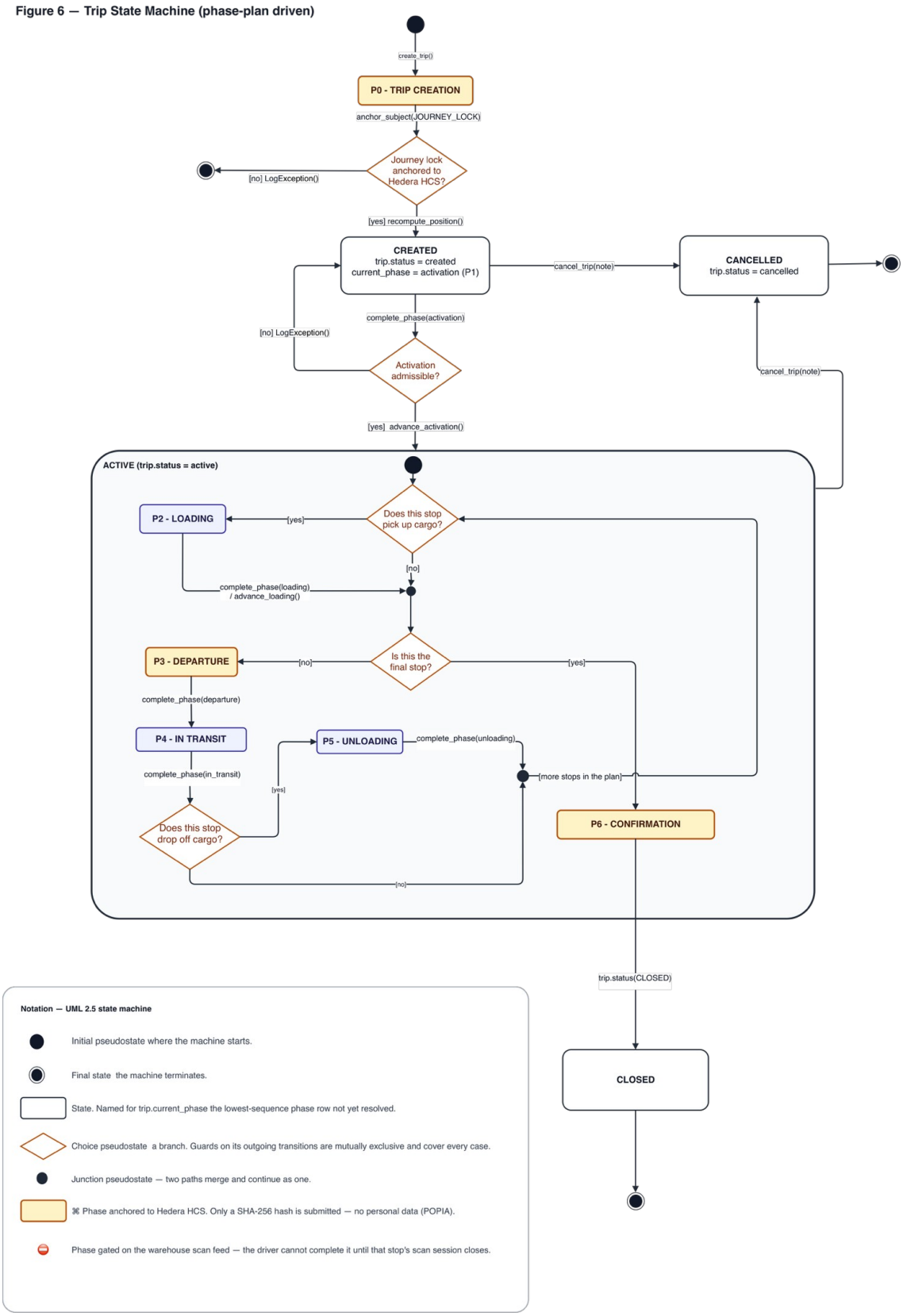


Figure 5: Trip state-machine diagram

TEAM TO DO: This is the Iteration 3 diagram, kept as a stand-in. Redraw it to match the four bullet points above: the refused-activation self-transition, [yes]/[no] guards on every decision, the per-stop order Arrival → Unloading → Loading, and the Arrival state.

Each exception also has a simple life cycle, enforced by BR-2: *Unreviewed* → *Claimed* (claim) → *Reviewed* (review). A claimed exception returns to *Unreviewed* when released, and a reviewed exception never changes state again.

6.7 Database Schema Changes

The relational schema is changed through versioned migrations only; CI applies the whole chain to an empty database on every change and checks that there is exactly one migration head. Table 21 lists the Iteration 4 changes. Tables that did not change are shown in the Iteration 3 entity relationship diagram (Team05, 2026b, Fig. 7, p. 39), and Figure 6 shows the diagram with the Iteration 4 tables.

Table 21: Schema changes in Iteration 4

Table	Change	Story
trips	Manifest key (issuer account, origin hub, manifest number), unique among non-cancelled trips; order number removed	FP-281
precincts	Parcel Perfect hub code, so a manifest's hubs resolve to precincts	FP-281
phase_events	Seal condition at arrival; seal-number entry method; manifest snapshot on the trip-creation row; arrival phase type	FP-270, FP-300
exceptions	Claiming dispatcher and claim time; review status for every severity	FP-280
exception_vehicles (new)	Links one exception to each affected horse or trailer	FP-297
evidence_anchor_events (new)	One row per anchored exception raise, review or cancellation, with its frozen payload and receipt	FP-290
audit_packs, audit_pack_access_events, audit_pack_email_codes (new)	Issued packs with their fingerprints, share-link hashes, access log and one-time codes	FP-279, FP-291
incident_declarations (new)	Notifications and declared facts for an incident (BR-3)	FP-279
cost_profiles (new)	Per-organisation cost rates for cost analytics	FP-294
driver_substitutions	Now written by the substitution workflow	FP-83
receiver_identity_verifications	Stored selfie reference before a "selfie only" result	FP-316

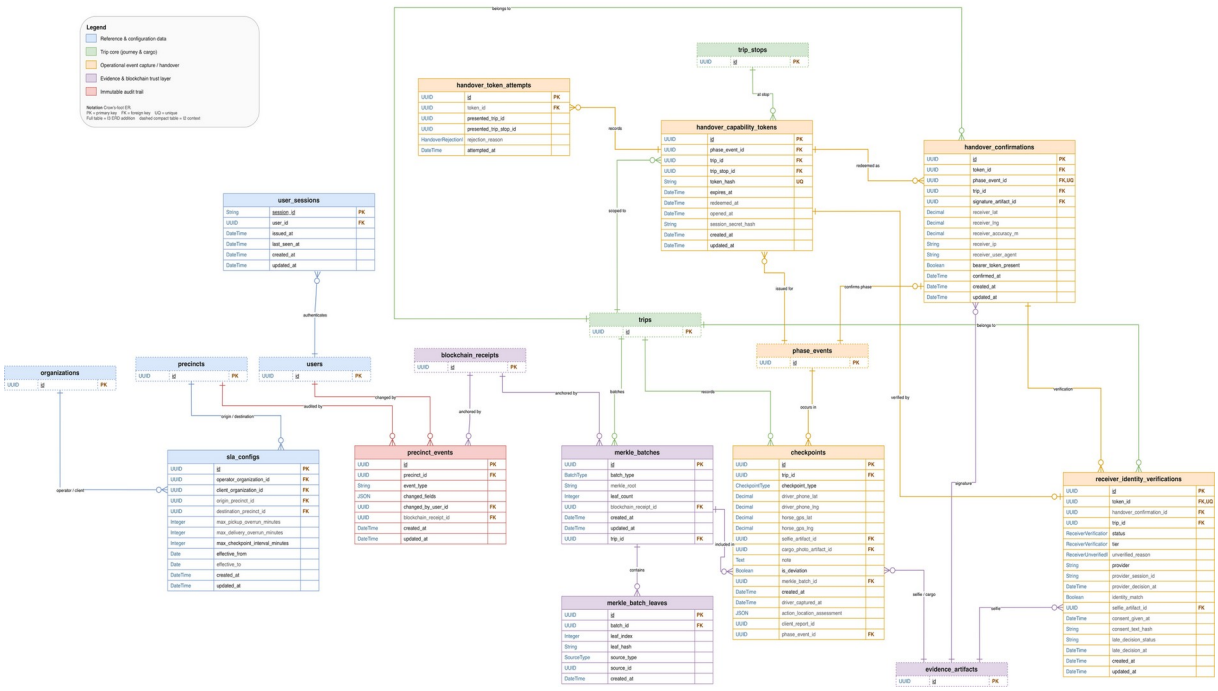


Figure 6: Entity relationship diagram

TEAM TO DO: This is the Iteration 3 ERD. Add the new tables and columns listed in the schema-changes table, highlighted in a different colour.

Every new table is created with row-level security enabled and direct access revoked, so it cannot be read through Supabase's data interface, following the database lockdown in FP-271. Personal data stays in PostgreSQL in af-south-1; only fingerprints are anchored.

7. Architecture

7.1 Deployment Architecture

The Iteration 3 architecture figure was criticised for looking like a package diagram. Figure 7 is therefore drawn as a deployment view: the devices people use, the services FreightProof runs, the external services it calls, and the protocol on each connection. The module structure is shown separately in the package diagram (§6.1). Table 22 lists the nodes.

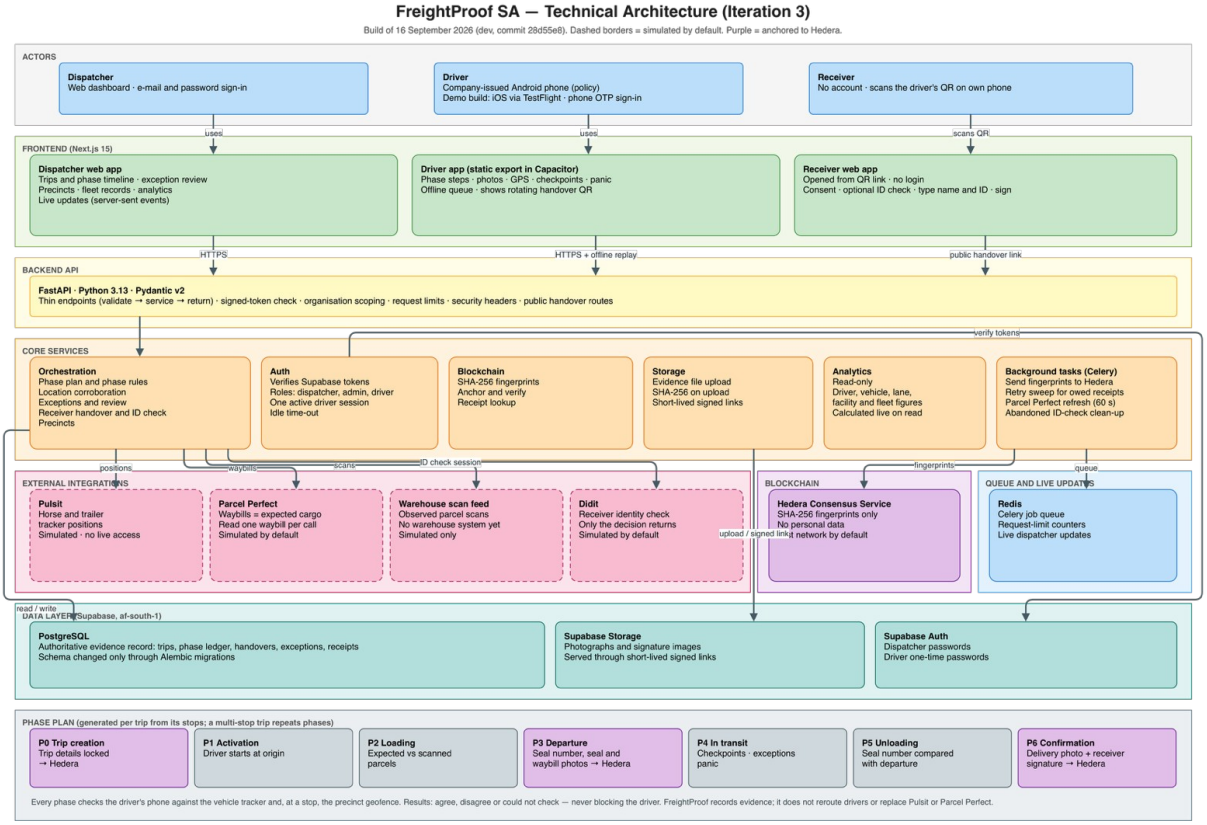


Figure 7: Deployment architecture diagram

TEAM TO DO: This is the Iteration 3 figure, which Ammar said looked like a package diagram. Redraw it as a deployment diagram matching the node table below: devices, Vercel, Railway, Supabase (af-south-1) and the external services, with the protocol on each connection.

Table 22: Deployment nodes and connections

Node	Runs on	Connections and purpose
Dispatcher application	Vercel (www.freightproof.co.za)	HTTPS/REST to the API; Server-Sent Events (SSE) for live trip and exception updates
Driver app	Driver's phone: iOS (TestFlight) and Android, in a Capacitor shell	HTTPS/REST to the API; actions queued on the phone when offline and sent when signal returns
Receiver application	Receiver's phone browser; Vercel (receiver.freightproof.co.za)	HTTPS to the API through a single-use link and a browser-binding cookie; Didit only when a check is required
Client portal	Recipient's browser; Vercel (portal.freightproof.co.za)	HTTPS to the API's public audit-pack endpoints; direct HTTPS to the Hedera mirror service for independent verification

Node	Runs on	Connections and purpose
API	Railway container (api.freightproof.co.za)	The only component that writes evidence; holds all business rules
Background worker and Redis	Railway	Anchoring jobs, retries and the recovery sweep; manifest refresh; rate limits; live-event fan-out
Supabase (af-south-1, Cape Town)	Managed PostgreSQL, file storage and authentication	Records, photographs, signatures and PDFs; dispatcher passwords and driver one-time passwords
External services	Hedera, Parcel Perfect, Pulsit, scan feed, Didit, SendGrid, OpenRouteService	Fingerprints out and receipts in; manifests and waybills; tracker positions; parcel scans; receiver checks; access-code e-mail; planned routes

Trust boundaries. Three boundaries matter. The receiver's browser and the recipient's browser are outside FreightProof's control, so what they submit is evidence, not proof. The API is the only writer of evidence. Hedera is outside FreightProof's control, which is exactly why it is used: an operator cannot rewrite a fingerprint after it has been published.

7.2 Integrity Model and Terminology

The Iteration 3 document used "fingerprinted" in a way that could be read as "digitally signed". The terms are now used precisely:

- A **fingerprint** is a SHA-256 hash of a record's committed values (National Institute of Standards and Technology [NIST], 2015). Any change to those values produces a different fingerprint. A fingerprint is not a digital signature: it shows *that* a record changed, not *who* created it.
- **Anchoring** publishes the fingerprint to a Hedera Consensus Service topic. Each submission is signed with FreightProof's Hedera account key, and Hedera's consensus assigns it an order and a timestamp (Hedera, n.d.). The receipt therefore proves that *this exact set of values existed no later than the consensus time*. It does not prove that the values were true.
- **Verification** rebuilds the record's committed values from the database, recalculates the fingerprint and compares it with the anchored one. The result is "matches", "does not match" or "could not check".

Iteration 4 extended anchoring from three phases to every phase (FP-270), and added anchored records for critical exceptions, every review and every cancellation (FP-290). Each action is a separate anchored event, so a later review never edits an earlier receipt. Free text, such as a description or review note, is committed only as its own SHA-256 hash, and positions are never committed, so no personal data reaches the public ledger.

7.3 Configured Values and Their Rationale

Iteration 3 stated several numbers without explaining them. Table 23 gives the reason for each. All are configuration settings, not constants in the code, and can be tuned in the pilot.

Table 23: Configured values and their rationale

Setting	Value	Why this value
QR code rotation	20 s	A photograph or screenshot of the code stops working within 20 s, which limits a forwarded or remote handover. Scanning a code that is on screen takes a few seconds, so honest use is not affected. The code stops rotating once the receiver opens it, and the driver can always show a fresh one.
Handover link lifetime	10 min	Long enough to complete a signature; extended once when an identity check starts, because a check can take several minutes.
Precinct geofence	200 m + 50 m tolerance (default)	A starting value that covers a typical depot yard. It is set per precinct (BR-11); 200 m was a sample in the Iteration 3 screenshots. The 50 m tolerance absorbs normal phone and tracker inaccuracy.
Driver-to-truck separation	100 m	A driver performing a handover stands at the truck. 100 m allows for a large yard and GPS error while still detecting a phone that is somewhere else.
Position freshness	60 s age; 30 s between phone and tracker	Comparing positions taken at different times would produce false mismatches. Older positions are "could not check".
Trailer-to-horse separation	500 m	A coupled trailer parked at the edge of a fence must not raise a critical alarm; a decoupled trailer on the road should.
Anchoring receipt target	60 s	Submissions take four to six seconds; 60 s leaves room for retries without delaying the driver.
Manifest refresh	60 s	Keeps expected cargo current while a trip is active, within Parcel Perfect's request limits.
Idle sign-out	10 min	Limits misuse of an unattended dispatcher session or phone.

7.4 Legal Standing of the Receiver's Signature

The Iteration 3 review asked whether a signature differs legally from an image of one. Under the Electronic Communications and Transactions Act 25 of 2002 (ECTA), a drawn signature captured on a phone is an ordinary *electronic signature*: data attached to a record and intended by the user to serve as a signature. Where the law requires a signature but the parties have not required a particular type, an electronic signature is valid if the method identifies the person, indicates their approval and is as reliable as is appropriate for the purpose (Electronic Communications and Transactions Act, 2002, s. 13). An *advanced* electronic signature, which needs an accredited provider and face-to-face identification, is only required in specific cases and is not used by FreightProof.

FreightProof's design therefore focuses on reliability rather than on the signature image itself. The confirmation binds the typed name, the drawn signature, the time, the receiver's position and the single-use link to one stop, and its fingerprint is anchored. This makes the confirmation strong evidence of who acknowledged the delivery and when, but it is not proof of identity unless the optional check was used. This is the team's reading of the Act, not legal advice; a legal opinion would be needed before a pilot.

7.5 Integration Status

Table 24 gives the final state of each integration.

Table 24: Integration status at the end of Iteration 4

Component	Implemented capability	Mode	Failure behaviour
Supabase	Records, files, authentication, organisation scoping, row-level security	Live	Requests refused; nothing written
Redis and worker	Anchoring, retries, recovery sweep, manifest refresh, live events	Live	Receipts owed and retried
Hedera	Fingerprints out, receipts in, mirror verification	Live (test network)	Phases complete and anchor later; trip creation refused
Parcel Perfect	Manifest import, waybill lookup and refresh	Simulated manifest feed	Trip creation refused
Scan feed	Scan gating and parcel comparison	Simulated	Phase waits; override under BR-4
Pulsit	Horse and trailer positions; on-road tracker check	Simulated	"Could not check"; driver continues
Didit	Opt-in receiver identity check	Live; off by default	Recorded as unverified; delivery continues
SendGrid	Audit-pack access codes	Live	Code not sent; dispatcher re-issues
OpenRouteService	Planned truck route and expected arrival	Live	Falls back to the planned arrival time

7.6 Quality Attributes

Security. Every request needs a signed token except the health check, the receiver pages, the public audit-pack pages and the identity provider's signed callback. Data is scoped to each organisation, fleet changes need an Admin-Dispatcher, and production refuses to start with unsafe settings. Iteration 4 locked down direct database access (FP-271), keyed rate limits on the verified user (FP-274) and enforced upload limits (FP-308).

Maintainability. CI runs linting, type checking, a function- and file-size limit, the layering check, the full test suite and a migration check on every pull request. Each external system sits behind an interface with simulated and live implementations.

Scalability and performance. Database connection and pool-wait timeouts make a slow database fail fast instead of hanging requests (FP-275). A repeatable load test against a separate environment measures these limits and anchoring latency (FP-299; §9.3). Analytics are calculated from read-only views each time they are read.

8. UI Mock-ups and Design Rationale

8.1 Design Approach

The Iteration 3 review asked the team to design screens for use, not for display: to decide what a user must see at first glance every time, what they look for only occasionally, and how the design changes when a company runs 10 or 50 trucks. Each screen in Iteration 4 was therefore designed from three questions: who uses it, how often, and what decision they make on it. Table 25 summarises the answers. The dispatcher application is used on desktops and tablets, so screens were designed and checked at 1,440, 1,280, 1,024 and 768 pixels wide; the driver and receiver applications are designed for phones.

Table 25: Screen design rationale

Screen	User	Used	Must be visible at first glance
Live trip list (§8.2)	Dispatcher, operations room	Continuously	Which trips need attention now: current phase, time since last update, unreviewed exceptions, location check
Trip timeline (§8.3)	Dispatcher	When a trip needs attention	What happened in each phase, in order, and what is wrong; exceptions inside the phase where they occurred
Exception inbox and review (§8.4)	Dispatcher	Several times a day	Severity, what happened, who has claimed it, what must be decided
Analytics: Today and Trends (§8.5)	Operations manager	Today: daily; Trends: weekly or monthly	Today: incidents on a map. Trends: where problems and cost concentrate, with the source of every figure
Receiver handover (§8.6)	Driver and receiver	At every delivery	One clear instruction per screen; nothing that slows a delivery
Precinct set-up (§8.7)	Admin-Dispatcher	Rarely	The geofence on a map, placed by address
Audit pack and portal (§8.8)	Dispatcher; insurer or principal	Per incident or claim	What the pack contains, what is anchored, and whether it verifies

8.2 Live Trip List: the Summary of the Summary

The Iteration 3 demonstration centred on one trip, but nobody watches a single 48-hour trip. The dispatcher's home page is now the live list of every running trip (FP-289). Each row shows the date, trip, manifest, driver and horse, and route, followed by the three things that decide whether to look closer: the **current phase and time since the last update**, the number of **unreviewed exceptions**, and the latest **location check**. Rows update live without refreshing, every column can be sorted, and a trip that needs attention is flagged. A dispatcher watching 50 trucks therefore reads the page in seconds and opens only the trips that need attention. The single-trip timeline (§8.3) is the drill-down.

INSERT SCREENSHOT HERE

Live trip list on the dispatcher home page (FP-289), with several active trips, at least one flagged for attention

Figure 8: Live trip list: every running trip at a glance

8.3 Trip Timeline

Each phase on the trip page is a card with a one-line verdict ("Checks passed", "Needs attention" or "In progress"), followed by the acts inside the phase as a live mini-timeline: what was recorded, then what was checked (FP-288). Photographs appear as soon as they are uploaded, not when the phase completes (FP-287). The design answers each of the Iteration 3 comments:

- **Exceptions are inside the phase where they happened**, not in a corner panel. A phase with an unreviewed exception turns amber and shows its count in the verdict line (Figure 9). Each exception opens the review panel directly.
- **Driver and vehicle details** appear once, in the trip summary above the timeline, with a call link to the driver. They are not repeated on each phase card, so the most visible space carries the verdict lines.
- **The current position on a multi-stop route** is explicit: completed phases show a tick, the current phase pulses and names its stop, and future phases are dashed and labelled "Not started" (Figure 10).
- **Order.** Phases run oldest to newest, because a trip is read as a sequence of custody events and later phases depend on earlier ones. The "Jump to current phase" button and the live list (§8.2) give the newest state immediately, so the dispatcher never scrolls to find it.

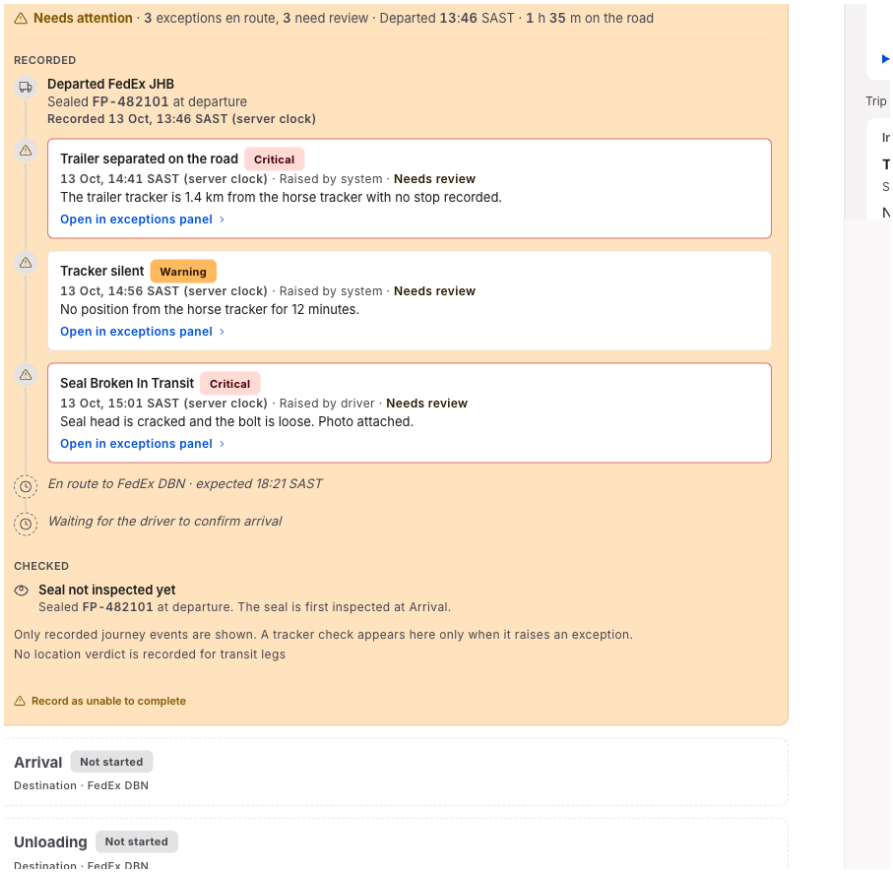


Figure 9: Trip timeline: in-transit phase with critical and warning exceptions

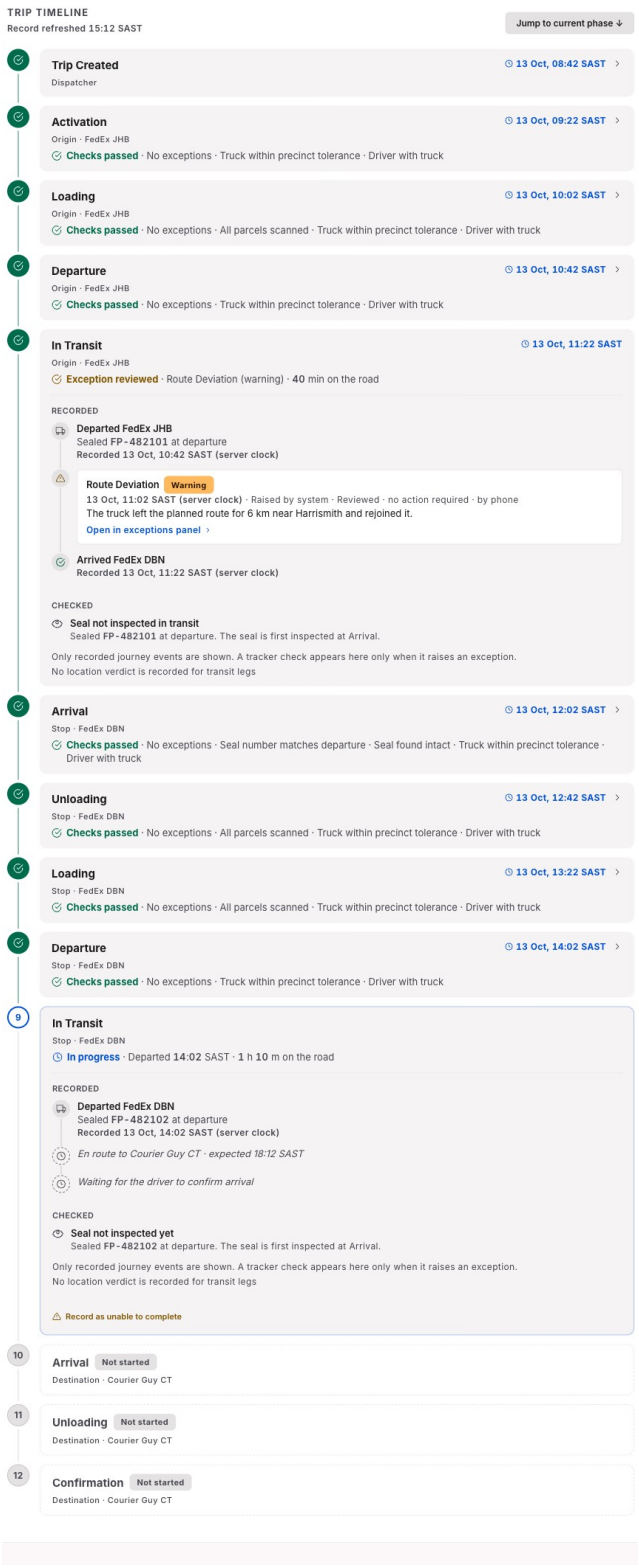


Figure 10: Trip timeline for a multi-stop trip, with the second leg in progress

8.4 Exception Inbox and Review

Every exception now reaches the dispatchers' shared inbox (FP-280), ordered by severity, with tabs for *Unreviewed* and *All recorded*. A dispatcher claims an exception before working on it, so two people do not phone the same driver, and the claim is visible to everyone live. The review form records the

outcome, the contact method (phone, WhatsApp or in person) and a note. Several warnings on one trip can be reviewed together, which keeps a demonstration trip with 15 minor findings from flooding the inbox. Reviewing never alters the original report; the review is anchored (FP-290).

The Iteration 3 review asked for exception handling "beyond typing". The process around the form is now defined by business rules: who must be informed and when (BR-3), which overrides need an Admin-Dispatcher's approval and supporting documents (BR-4), and how a later inspection result is recorded as an incident declaration rather than an edit (BR-5). Notifications to the principal, the insurer and SAPS are recorded as incident declarations with their time and reference numbers, and appear in the audit pack. Figure 11 shows the exception inbox and review form.

INSERT SCREENSHOT HERE

Exception inbox (FP-280) showing the Unreviewed tab, one exception claimed by a dispatcher, and the review form with outcome and contact method. Replaces the Iteration 3 exception-detail screenshot

Figure 11: Exception inbox with claims, and the review form

8.5 Analytics: Today and Trends

Analytics were split by how often they are used. The **Today** view is the operations-room screen: today's incidents on a map, which update by themselves when a report is raised and can be opened directly from a link (FP-296). The **Trends** view is for weekly and monthly analysis: activity, on-time performance, problems, review workload, evidence, routes and sites, and a new **Cost** tab (FP-294). The Cost tab shows the declared value of cargo on trips that had a theft signal, and estimated operating and delay costs from rates an Admin-Dispatcher can edit.

Two rules make the figures defensible. Every chart states its **source**, including when a figure depends on the simulated Pulsit or Parcel Perfect feeds, and figures based on fewer than five trips are flagged as a small sample. Analytics describe patterns that need investigation; they are not driver scores. An **AI insights brief** summarises patterns from de-identified figures only, such as which lane and time of day had the most in-transit exceptions in the period, and names the figures each statement rests on (FP-295). Figure 12 and Figure 13 show the Today view and the Cost tab; Figure 14, Figure 15 and Figure 16 show the headline tiles, the incident map and a vehicle's analytics.

INSERT SCREENSHOT HERE

Analytics Today view: today's incidents on the map with a pin raised during the demo trip (FP-296)

Figure 12: Analytics Today view: today's incidents on the map

INSERT SCREENSHOT HERE

Analytics Cost tab: cargo value exposed and estimated trip and delay costs, with the Source line visible (FP-294)

Figure 13: Analytics Cost tab

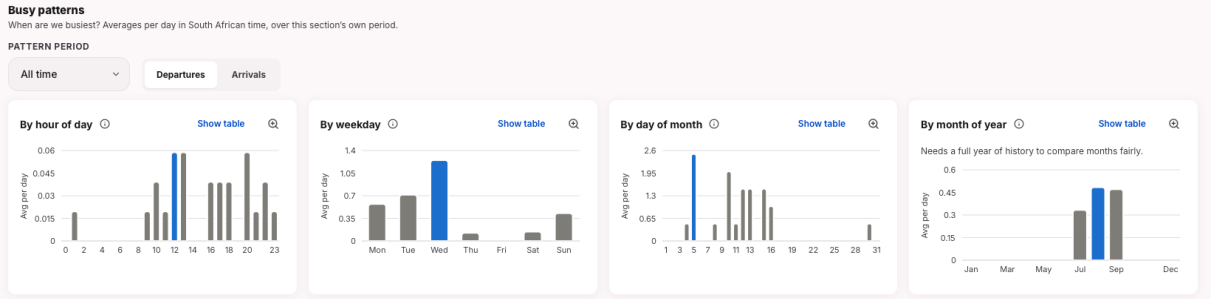


Figure 14: Analytics headline tiles

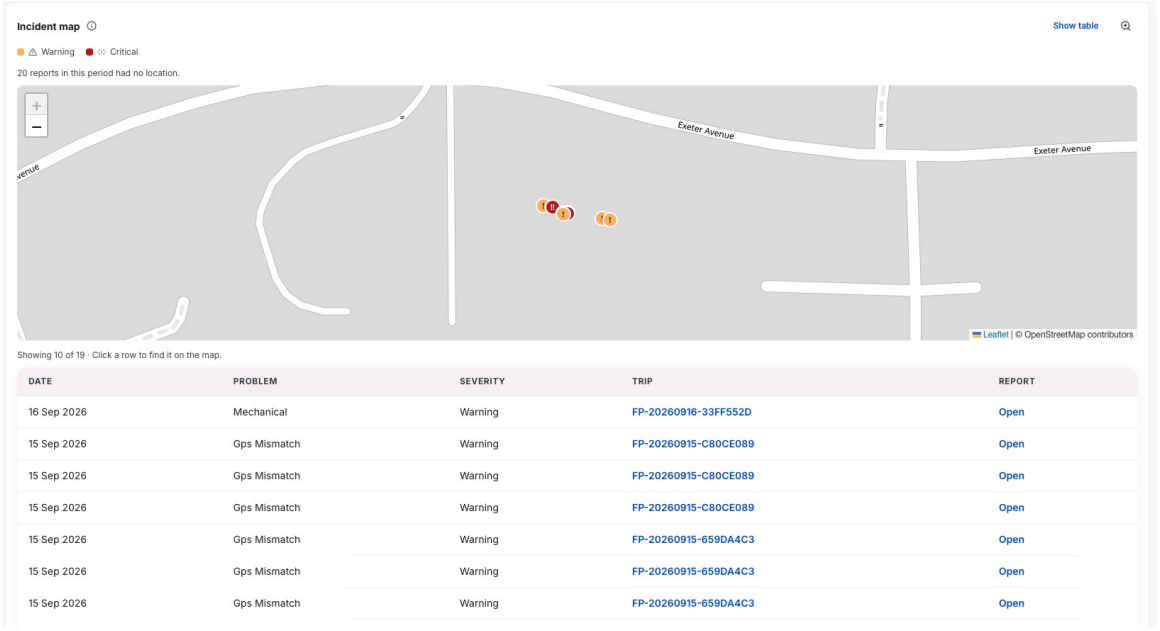


Figure 15: Incident map in the Today view

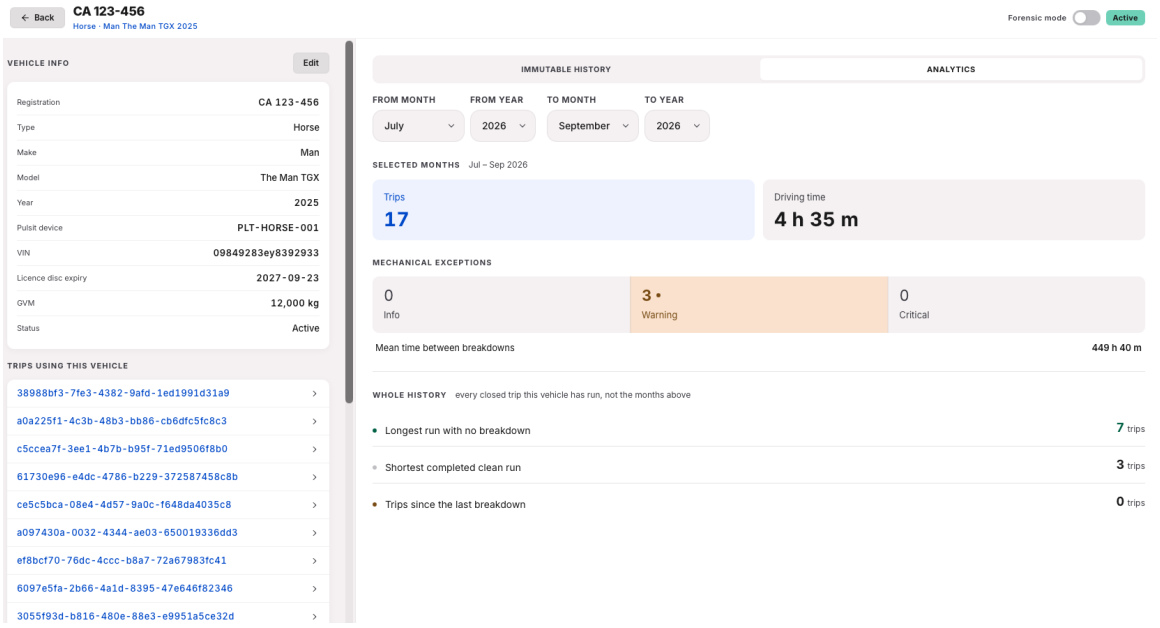


Figure 16: Analytics for one vehicle

TEAM TO DO: The tiles, incident map and vehicle analytics above are Iteration 3 screenshots. Retake them from the current build so that each chart's Source line is visible.

8.6 Driver App: Arrival and Receiver Handover

At Arrival the driver photographs the seal; the app reads the seal number from the photograph and asks the driver to confirm or correct it, instead of typing it (FP-300). The seal's condition is recorded as intact, damaged or missing (Figure 17).

INSERT SCREENSHOT HERE

Driver app Arrival screen: seal photograph with the number read from it and the confirm / correct buttons (FP-300)

Figure 17: Driver app: seal number read from the photograph at Arrival

The driver shows a QR code that rotates every 20 seconds (Figure 18). The receiver scans it with their own phone, types their name, signs and confirms; an identity check appears only for consignments that require it (BR-7). The driver's screen then shows that the receiver confirmed, at what time, and from their own device (Figure 19). The Iteration 3 receiver screenshots showed a real identity document and have been withdrawn; the screens shown here contain no personal data.

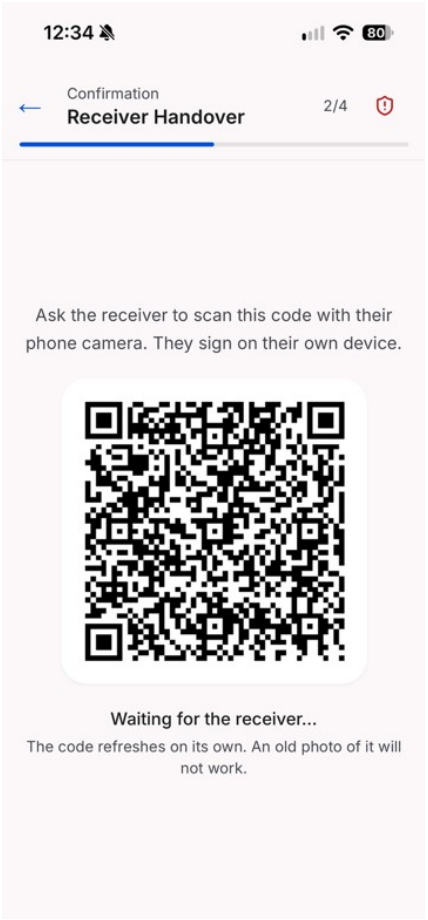


Figure 18: Driver app: QR code for the receiver

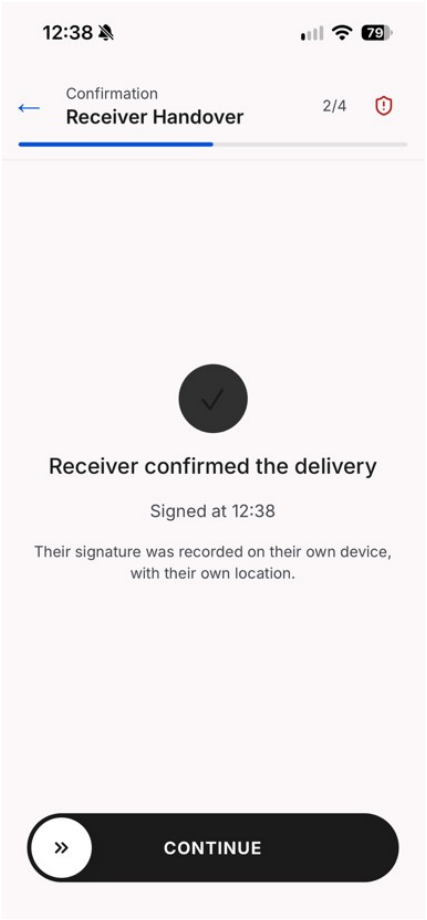


Figure 19: Driver app: receiver confirmation received

8.7 Precinct Set-up

An Admin-Dispatcher now finds a precinct by searching for its address, which places the pin on the map and fills in the coordinates (FP-309); a click on the map shows the address of that point. The geofence radius is drawn on the map and set per precinct (BR-11). Figure 20 shows the precinct list with each geofence and whether the precinct is owned or shared, and Figure 21 the address search.

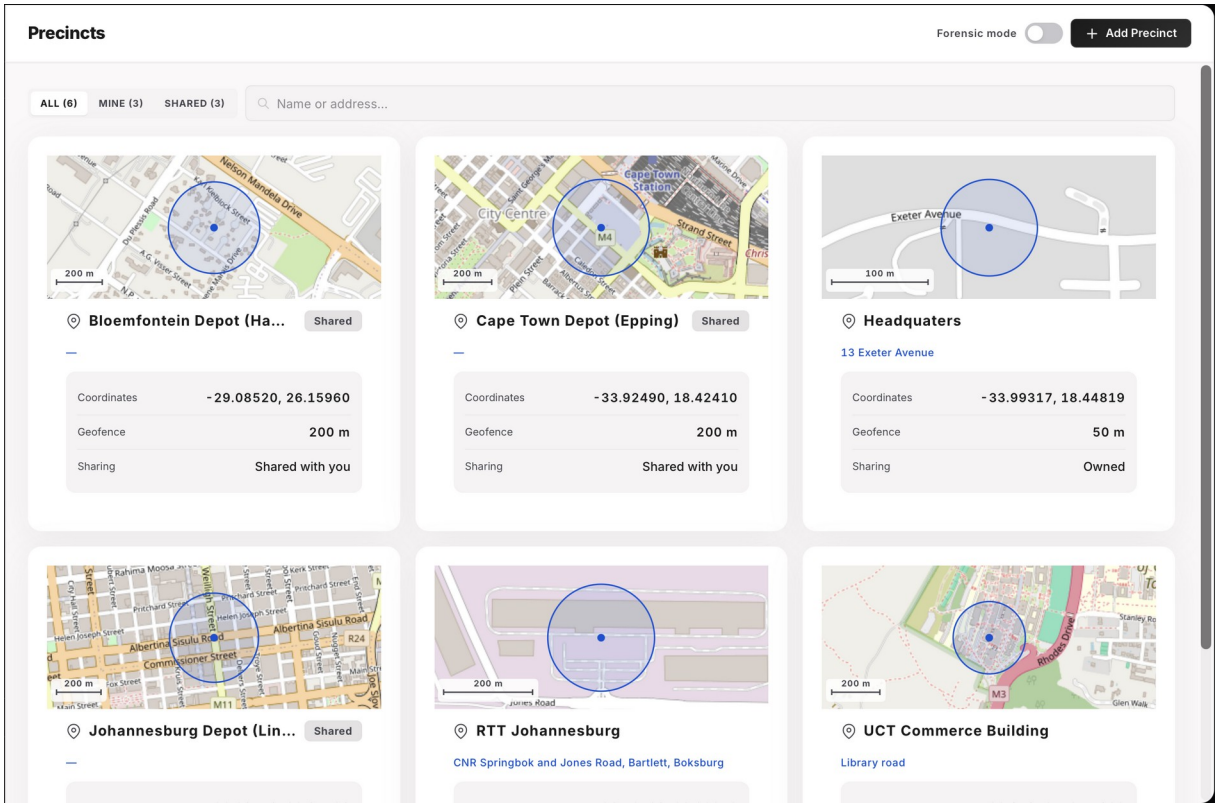


Figure 20: Precinct list with geofences

INSERT SCREENSHOT HERE

Precinct form with an address typed into the search box and the pin and geofence placed on the map (FP-309)

Figure 21: Precinct set-up by address search

8.8 Audit Pack and Client Portal

A new *Audit packs* tab on the trip page (Figure 22) lets a dispatcher preview a trip's audit trail, issue a pack to a named recipient, see each pack's status and access log, and revoke it (FP-279). The recipient's page in the client portal (Figure 23) presents the same content as the PDF: a timeline, a route map, photographs, exceptions, the parties involved, and an integrity section with a **Verify** button. Every item is labelled as *anchored* or *recorded only*, and missing evidence is listed rather than hidden, because an insurer must know what the pack does not prove.

INSERT SCREENSHOT HERE

Dispatcher trip page, Audit packs tab: list of issued packs with status, access log and the Issue button (FP-279)

Figure 22: Audit packs tab on the trip page

INSERT SCREENSHOT HERE

Client portal: an audit pack opened by a recipient, with the integrity section showing the result of Verify (FP-279, FP-291)

Figure 23: Client portal: verifying an audit pack

9. Testing Approach, Results and Plan

The testing approach is unchanged in principle from Iteration 3: automated unit and integration tests organised around the project risks, concurrency tests that hold two connections at a barrier so that they genuinely collide (Team05, 2026b, §7.1.1, pp. 57–58), and story acceptance walkthroughs. Iteration 4 added three things the Iteration 3 review asked for: the full test suite now runs automatically on every change, failure scenarios (lost signal, dead phones, cleared app data and outages) are tested deliberately, and load is measured.

9.1 Kinds of Tests

Table 26: Test levels used in Iteration 4

Level	Tool and scope	Iteration 4 focus
Backend unit	pytest; one piece of logic, no database or network	Phase-plan generation with Arrival; anchored payloads; corroboration limits; severity; cost calculations
Backend integration	pytest and httpx against a real PostgreSQL database and Redis	Manifest trips; Arrival; exception claims and batch review; anchored exceptions, reviews and cancellations; audit packs; parcel trace; vehicle clashes
Frontend unit and component	Vitest; screens, hooks and client logic in all four front ends	Offline queue and exception store; timeline phase cards; exception inbox; audit-pack panel; portal verification
Concurrency	Barrier-synchronised database connections	Two claims or reviews of one exception; two redemptions of one QR link; two trips for one manifest; simultaneous cancellations
Security	Integration tests per role and organisation	Organisation isolation, identical "not found" responses, locked-down database tables, per-user rate limits
Static and structural	ruff, mypy, import-linter, size limits	Layering rule enforced; no oversized functions or files
Migration	Alembic against an empty database in CI	Exactly one migration head; the whole chain applies cleanly
Load	Scripted concurrent clients against a separate environment (FP-299)	Concurrent phase completions and dashboard reads; database timeouts; anchoring latency
Device and connectivity	Manual, on Android and iPhone handsets	Airplane mode mid-phase, battery dies, app closed, app data cleared, API or partner service down
Story acceptance	Team walkthroughs on the deployed system	Each Sprint 8 and 9 story against its acceptance criteria

9.2 Automated Test Evidence

The repository contains 227 backend test files (109 unit and 112 integration) and 362 frontend test files across the dispatcher, driver, receiver and client-portal applications. Figure 24 shows the result of running the suites on the development branch on 19 October 2026: 3,411 backend tests, 1,448 driver-app tests and 2,900 dispatcher tests passed, with no failures. The 267 deselected backend tests are marked slow and are excluded from this run, as they are in the main CI step. Since Iteration 4, CI runs

the complete backend suite on every pull request; in Iteration 3 it stopped at a style check (Team05, 2026b, p. 43). Table 27 lists the CI steps, all of which must pass before a change can be merged into the protected development branch.

```
$ cd backend && pytest -m "not slow"   # PostgreSQL + Redis, dev branch, 19 Oct 2026
3411 passed, 4 skipped, 267 deselected, 1971 warnings in 281.70s (0:04:41)

$ cd frontend/driver-pwa && npx vitest run
 Test Files  146 passed (146)
      Tests  1448 passed (1448)
   Duration  181.42s

$ cd frontend/dispatcher && npx vitest run
 Test Files  202 passed (202)
      Tests  2900 passed (2900)
   Duration  273.33s
```

Figure 24: Summary output of the backend, driver-app and dispatcher test suites, 19 October 2026

Table 27: Continuous integration pipeline, run on every pull request

Step	What it checks
Lint (ruff)	Code style and common errors
Structure ratchet	No function or file grows beyond the agreed size limits
Layering (import-linter)	No module skips a layer, for example a request handler calling an integration directly
Type check (mypy)	Type consistency across the backend
Tests (pytest)	Full unit and integration suite against a throwaway PostgreSQL 17 database and Redis
Import check	Every module imports cleanly in a fresh interpreter, with no import cycles
Migrations	Exactly one migration head; the whole chain applies to an empty database
Frontend	ESLint, TypeScript type check and Vitest suites for the dispatcher, driver and receiver applications

9.3 Failure-Scenario and Load Testing

The Iteration 3 review asked for serious testing of location, connectivity and outages. Table 28 lists the scenarios tested and the expected behaviour, which is defined by the business rules in §6.4. Each scenario is also a test case in Appendix A where an automated test covers it.

Table 28: Failure scenarios and expected behaviour

Scenario	Expected behaviour
No signal during a phase	The action is queued on the phone and sent when signal returns; the server records it once, using the same identifier on every retry
A queued action is blocked because an earlier phase is incomplete	The item is kept and retried, not discarded (FP-276); the dispatcher is alerted if it cannot be delivered
App data or cache cleared before sync	Unsent captures are lost on the phone, but nothing false is recorded: the phase stays pending on the server and on the live trip list, and the driver recaptures it

Scenario	Expected behaviour
Battery dies, phone lost or stolen	The phase stays pending; a sign-in on another phone ends the old session; overrides only under BR-4 (BR-6)
Tracker silent or position stale	"Could not check" is recorded, never a mismatch; the driver continues
Hedera unavailable	Phases complete and their receipts are marked owed and retried; trip creation is refused
Parcel Perfect unavailable	Trip creation is refused; an active trip keeps its last manifest
Didit unavailable	The receiver's check is recorded as unverified with a reason; delivery continues
API unavailable	The driver app queues actions and retries; the dispatcher application shows the failure rather than stale data
Slow database	Connection and pool-wait timeouts end the request quickly with an error, rather than hanging (FP-275)

The load test (FP-299) runs concurrent drivers completing phases, location updates and dashboard reads against a separate environment, never the shared development database. Its success criteria are no server errors or duplicate records under sustained load, database timeouts that fail fast, and 95% of anchor receipts within 60 seconds.

TEAM TO DO: Add the load-test results here: **[number of simulated drivers and duration]**, **[errors and duplicate records]**, **[95th-percentile anchoring time]**, **[slowest database request under load]**.

9.4 Risk-to-Test Matrix

Table 29 links each project risk to the tests that cover it. Layer B is backend and F is frontend.

Table 29: Risk-to-test matrix

Risk	What is tested	Layer	Test cases
R-02, R-04	Every phase, critical exception, review and cancellation anchored; each action has its own receipt; tampering detected	B	TC-04, TC-05
R-03	Phone, horse and trailer compared at every phase; a silent tracker gives "could not check"	B	TC-06
R-05	Offline queue keeps blocked items and replays each item once	F, B	TC-12
R-06	Anchored payloads contain no positions, names or free text	B	TC-05
R-13	One redemption per link; a second browser cannot confirm; rotation pauses once the link is opened	B	TC-08, TC-09
R-12	Identity check only when the consignment requires it; outcome recorded	B	TC-10
R-14	Organisation isolation; locked-down tables; per-user rate limits	B	TC-11
R-16	Full suite, layering and migrations in CI	B, F	§9.2
SO-3	Manifest trip creation, duplicates refused, parcel trace	B	TC-01, TC-02, TC-03
SO-5	Every exception reviewed; claims; concurrent reviews	B	TC-07
SO-6	Seal mismatch, damage and absence at Arrival	B	TC-06
SO-8	Audit pack issue, access code, revocation, verification	B, F	TC-13, TC-14

9.5 Test Plan and Status

Table 30 updates the Iteration 3 test plan (Team05, 2026b, Table 37, p. 60).

Table 30: Test plan and status at the end of Iteration 4

What	Environment	Status
Load and volume, including anchoring latency	Separate load-test environment	Done (FP-299); results in §9.3
Live identity verification and callback	Deployed system with the live provider	Done; the check is now opt-in per consignment
Live Parcel Perfect and Pulsit integration	Partner sandbox	Not possible: credentials not granted (R-22). Simulated feeds tested instead
Override and driver completion of the same phase at the same instant	Backend integration	Done: exactly one write succeeds
Device and connectivity scenarios (Table 28)	Android and iPhone handsets on the deployed system	**[TO CONFIRM: date run, handsets used and result]**
Acceptance testing with LFG users	Deployed system	Not done within the project; team story acceptance only. Planned as the first step of the pilot (BO-1)

10. Development Approach, Scrum and Progress

10.1 Development Approach and Standards

Team 05 developed FreightProof iteratively with Scrum, tracked on the Jira board FP. Roles were unchanged throughout the project: Ciaran Formby was Product Owner, Tim Gultig was Scrum Master, and all four members, with Thomas Davis and Chiko Kasongo, formed the Development Team. Each sprint opened with planning, in which every story was estimated in Fibonacci story points and assigned before work started. The team met in full every Wednesday, coordinated dependent work in the team channels, and closed each sprint with a review against Jira and a retrospective. Each iteration ended with a panel review, whose feedback became the next iteration's backlog.

The team followed these standards:

- **Version control.** The main and development branches are protected. Every change arrives through a pull request that a second member reviews, and commits reference their Jira key.
- **Definition of done.** A story is done when its acceptance criteria are met, its tests pass in CI, it has been reviewed and merged, and its Jira ticket links the pull request.
- **Coding standards.** The Python backend uses typed code, Pydantic v2 models and async database access, checked by ruff and mypy. The four front ends use TypeScript in strict mode with ESLint. The layering rule is checked by CI (§9.2).
- **Database standards.** Schema changes go through named, hand-reviewed migrations only. Every new table has row-level security enabled.
- **UI standards.** A shared design system (colours, type, spacing and status chips) is used by all front ends, and shared types come from one package so that a contract change fails at compile time.

10.2 Sprint 8

Sprint 8, 29 September to 12 October 2026 (13 days). **Goal:** complete the final features and begin preparing for final submission. Table 31 lists the committed backlog, written as user stories.

Table 31: Sprint 8 committed backlog (19 items, 84 points)

Key	User story	Assignee	Pts	Outcome
FP-281	As a dispatcher, I want to create a trip from the client's manifest number, so that cargo is imported, not typed	Ciaran Formby	11	Done in sprint
FP-280	As a dispatcher, I want every exception to reach a shared inbox where I can claim it, so that nothing is recorded and forgotten	Ciaran Formby	8	Done in sprint
FP-149	As a dispatcher, I want to find a parcel by barcode and see everywhere it has been	Chiko Kasongo	8	Done in sprint (carried from Sprint 7)
FP-279	As an insurer, I want an anchored audit pack that I can verify myself, so that I need not trust the operator's copy	Tim Gultig	8	Done in sprint
FP-270	As a dispatcher, I want the seal inspected at Arrival and every phase anchored, so that a break in transit is detected when custody arrives	Ciaran Formby	3	Done in sprint
FP-272	As a demonstrator, I want a demo panel that follows the trip, so that partner events can be simulated in order	Ciaran Formby	2	Done in sprint

Key	User story	Assignee	Pts	Outcome
FP-277	As a developer, I want the identity-provider call inside the service layer, so that request handlers never call an integration directly	Tim Gultig	2	Done in sprint
FP-271	As an operator, I want newer tables closed to direct database access, so that evidence cannot be read around the API	Ciaran Formby	1	Done in sprint
FP-295	As an operations manager, I want a plain-language brief of patterns in my analytics, based on de-identified figures	Thomas Davis	8	Carried; done in Sprint 9
FP-286	As the team, we want documentation and evidence captured during the sprint, so that the final document is assembly, not authoring	Unassigned	5	Carried; done in Sprint 9
FP-294	As an operations manager, I want to see the cargo value exposed and the estimated cost of delays	Thomas Davis	5	Carried; done in Sprint 9
FP-296	As an operations manager, I want today's incidents on a map and the source of every figure	Thomas Davis	5	Carried; done in Sprint 9
FP-300	As a driver, I want the seal number read from my photo, so that I confirm it instead of typing it	Tim Gultig	5	Carried; done in Sprint 9
FP-273	As a principal, I want delivery to close only after the receiver confirms, so that a driver cannot sign for the receiver	Tim Gultig	3	Carried; done in Sprint 9
FP-276	As a driver, I want evidence captured offline to be kept until it is accepted, not discarded	Tim Gultig	3	Carried; done in Sprint 9
FP-274	As an operator, I want request limits applied per user, so that one user cannot exhaust another's	Chiko Kasongo	2	Carried; done in Sprint 9
FP-275	As a dispatcher, I want a slow database to fail fast rather than freeze my screen	Chiko Kasongo	2	Carried; done in Sprint 9
FP-297	As a driver, I want to report a breakdown that affects both the horse and a trailer	Ciaran Formby	2	Carried; done in Sprint 9
FP-278	As a driver, I want real seal numbers accepted, not only an invented format	Tim Gultig	1	Carried; done in Sprint 9

Outcome. Sprint 8 completed 43 of 84 points (51%) within the sprint and carried 41 points into Sprint 9; no scope was dropped. Figure 25 shows the burndown, drawn from the Jira resolution dates. The scope line stays at 84 points throughout because no stories were added after planning; the remaining work therefore never rises. It falls in steps rather than steadily because large stories close in one step: the 11-point manifest story and the 8-point exception inbox closed on 5 and 6 October.

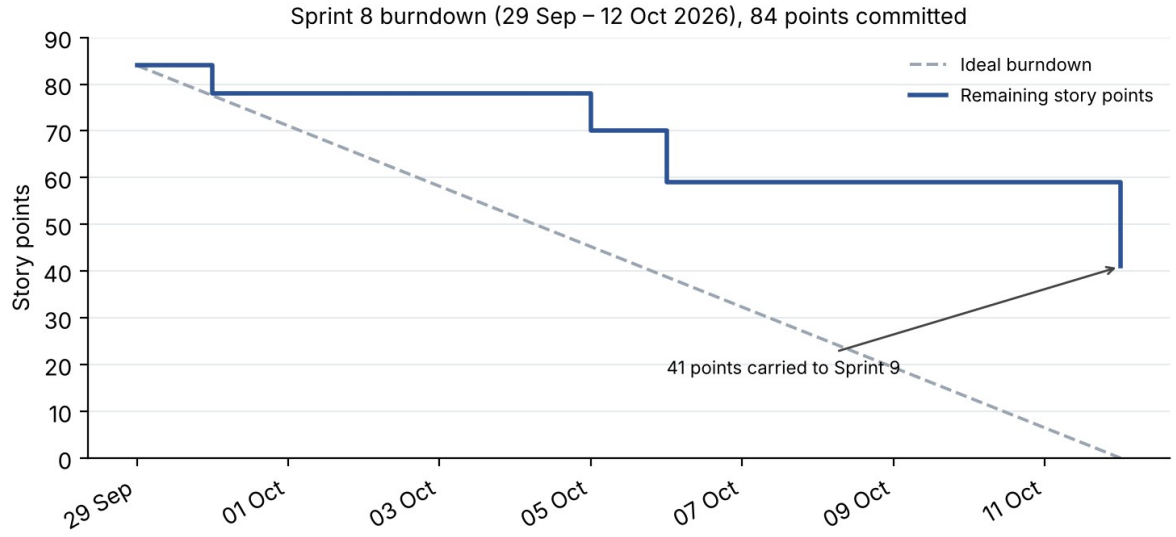


Figure 25: Sprint 8 burndown chart

INSERT SCREENSHOT HERE

Jira board FP, Sprint 8 task board (or sprint report) at sprint close, 12 October

Figure 26: Sprint 8 task board

TEAM TO DO: The Sprint 8 and Sprint 9 retrospectives below were drafted from the Jira data. Confirm or rewrite them with what the team actually discussed.

Table 32: Sprint 8 retrospective

What went well	What should improve	Action taken
The panel's feedback was turned into sized stories before the sprint started, and the largest items (manifest trips, exception inbox, audit packs) were delivered.	Only 51% of the commitment was completed within the sprint; several large stories were started late and finished early in Sprint 9.	Sprint 9 started with the carried stories, and carried work was listed separately from new work.
Carried items were reported separately from new work, as promised in Iteration 3.	Some points closed on 30 September for work already built on a branch before the sprint started, which overstated early progress.	Work is ticketed before it starts, even between sprints.
CI caught migration and layering problems before they reached the shared branch.	The documentation ticket had no owner and slipped again.	Documentation tasks were given owners and points in Sprint 9 (FP-303).

10.3 Sprint 9

Sprint 9, 12 to 23 October 2026 (11 days, including the code freeze from 19 October). **Goal:** close the Iteration 3 feedback, harden the system and deliver the final submission. Sprint 9 committed 118 points: the 41 points carried from Sprint 8 (Table 31) and 77 points of new work (Table 33).

Table 33: Sprint 9 new backlog (15 items, 77 points)

Key	User story	Assignee	Pts	Outcome
FP-303	As the team, we want the final document, evidence and code packaged for submission	Ciaran Formby	13	Done
FP-288	As a dispatcher, I want each phase on the timeline to show its acts live, so that I see what happened inside it	Ciaran Formby	8	Done
FP-290	As an insurer, I want critical exceptions, reviews and cancellations anchored, so that the records a dispute questions are protected	Ciaran Formby	8	Done
FP-308	As an operator, I want upload limits, artefact ownership and session checks enforced	Tim Gultig	8	Done
FP-287	As a dispatcher, I want photographs to appear on the timeline as soon as they are uploaded	Ciaran Formby	5	Done
FP-289	As a dispatcher, I want the trip list to show each trip's phase, last update, unreviewed exceptions and location check	Ciaran Formby	5	Done
FP-139	As a dispatcher, I want a horse that is on a live trip to be refused for a second one	Thomas Davis	5	Done
FP-309	As an Admin-Dispatcher, I want to place a precinct by searching for its address	Tim Gultig	5	Done

Key	User story	Assignee	Pts	Outcome
FP-83	As a dispatcher, I want a driver substitution recorded by both drivers, so that each driver's custody stays on the record	Ciaran Formby	5	Done
FP-142	As a dispatcher, I want to see a vehicle clash before I create a trip	Thomas Davis	3	Done
FP-291	As an operator, I want audit-pack links to need an e-mailed code before first view	Tim Gultig	3	Done
FP-316	As an insurer, I want a "selfie only" verification to exist only when the selfie is stored	Tim Gultig	3	Done
FP-292	As an insurer, I want the client portal deployed, so that a share link works outside the team's machines	Tim Gultig	2	Done
FP-293	As a receiver, I want an identity check only when the consignment requires it	Chiko Kasongo	2	Done
FP-299	As the team, we want a load test that proves the database timeouts hold	Chiko Kasongo	2	Done

INSERT SCREENSHOT HERE

Jira board FP, Sprint 9 task board at sprint close, 23 October

Figure 27: Sprint 9 task board

INSERT SCREENSHOT HERE

Jira burndown chart for Sprint 9 (Reports → Burndown chart), with any scope changes visible

Figure 28: Sprint 9 burndown chart

Sprint 9 retrospective. *Went well:* the panel's two central requests, a multi-trip view and anchored incident evidence, were delivered, and the code freeze held. *Should improve:* the sprint carried 118 points, which only worked because 41 were already partly built; workload remained uneven, as Table 35 shows. *Action:* recorded in the project reflection below (§10.5).

10.4 Velocity and Capacity

Figure 29 shows committed and completed points for every sprint. Sprint lengths varied between 10 and 14 days because sprints were fitted around the iteration reviews and the submission date. To compare them, Table 34 also gives points per sprint day. Jira was used sparsely in Sprints 1, 3 and 4 (4, 7 and 4 issues), so their figures understate the work done and are not comparable.

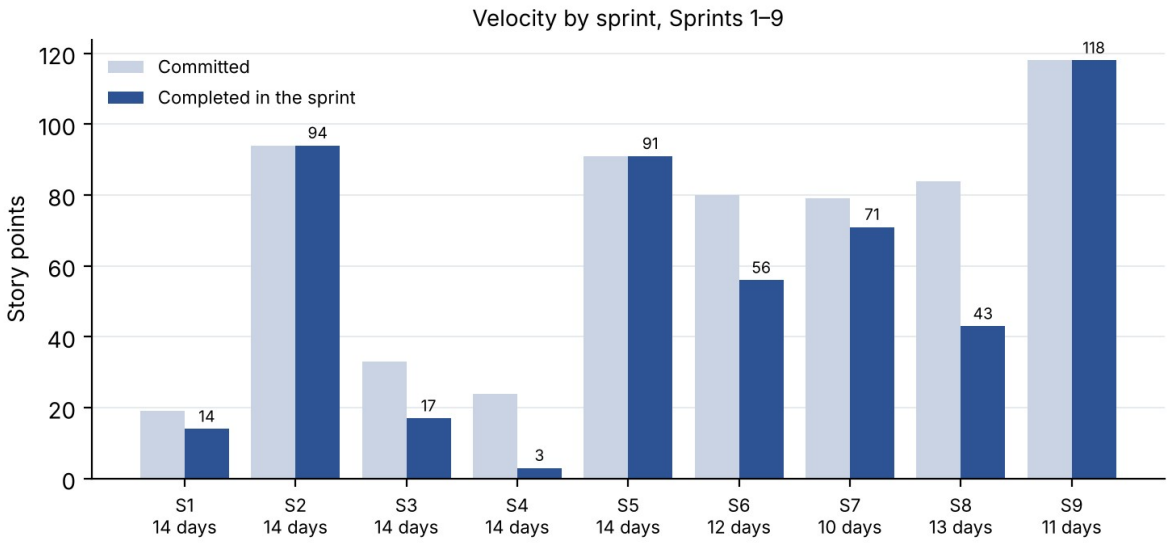


Figure 29: Committed and completed story points, Sprints 1–9

Table 34: Velocity normalised for sprint length

Sprint	Dates	Days	Committed	Completed	Points per day
2	3–17 May	14	94	94	6.7
5	27 Jul – 10 Aug	14	91	91	6.5
6	26 Aug – 7 Sep	12	80	56	4.7
7	7–17 Sep	10	79	71	7.1
8	29 Sep – 12 Oct	13	84	43	3.3
9	12–23 Oct	11	118	118	10.7

Across comparable sprints the team completed about 6.5 points per sprint day. Sprint 8 fell well below this and Sprint 9 well above it, because 41 points of Sprint 8 work were finished early in Sprint 9. Taken together, Sprints 8 and 9 delivered 161 points in 24 days, 6.7 points a day, which matches the team's earlier rate.

Table 35: Story points completed per member in Iteration 4

Member	Sprint 8	Sprint 9	Iteration 4	Share
Ciaran Formby	25	46	71	44%
Tim Gultig	10	33	43	27%
Thomas Davis	0	26	26	16%
Chiko Kasongo	8	8	16	10%
Unassigned (documentation)	0	5	5	3%
Total	**43**	**118**	**161**	**100%**

Story points measure feature tickets, not all contributions: documentation, research interviews, meeting minutes, testing and presentation work are largely outside them. Contributions in those areas are described in the portfolio of evidence (Appendix C).

10.5 Reflection on Scrum and Lessons Learnt

How Scrum helped. Scrum's short cycles let the panel's feedback change the product each iteration. Iteration 2's review led to the plan-driven phase lifecycle; Iteration 3's led to location corroboration and the receiver's own confirmation; and Iteration 4's led to the live trip list, the exception inbox and the audit pack. Without fixed review points, these changes would have arrived too late to build. The rule that a story must be estimated and assigned before it starts, adopted after Iteration 2, made planning reliable from Sprint 6 onwards. Pull-request reviews and CI made four people working on one codebase safe, and Jira gave an honest record of what was and was not finished.

Lessons learnt.

1. **Keep sprints the same length.** Varying lengths made velocity hard to compare and planning harder. A fixed two-week sprint, with reviews placed between sprints, would have been simpler.
2. **Treat documentation as sprint work.** Documentation was under-counted in three iterations and done at the end. It only improved when it became pointed tickets with owners.
3. **Write smaller stories.** Stories of 8 to 13 points closed in one step at the end of a sprint, which hid progress and made the burndown uninformative.
4. **Balance the workload deliberately.** Points were concentrated in two members in every iteration despite being raised in each retrospective. It has to be addressed at sprint planning, when work is assigned, not only noted in the retrospective.
5. **Ticket work before doing it.** Work done between sprints and later credited to a sprint distorted the sprint figures.
6. **Review AI-assisted code as carefully as your own.** Code that works is not the same as code the team understands (§10.6).
7. **Talk to users early.** The seven interviews and the meetings with LFG changed the product more than any technical decision: they produced the audit pack, the exception review workflow and the manifest-first trip.

10.6 Use of AI Tools

The team used AI tools throughout the project. Claude Code was the main tool, used to understand unfamiliar code, generate and refactor code, write tests, and draft design notes and specifications. Gemini was used to reconcile sprint records against the repository. AI was also used to check documentation against the code and to find merge conflicts across branches.

The team controlled this use in four ways. Repository instructions require a written plan before any non-trivial change and forbid AI tools from committing or merging. Every change passes through a pull request reviewed by a team member. CI enforces the tests, types and layering rule regardless of who wrote the code. Each design note records which decisions a team member made and signed off. The Iteration 3 code review showed that working code is not the same as understood code, so in Iteration 4 each member traces the data flow of their own modules ahead of the final code review on 28 October.

In the product, the AI insights brief (FP-295) summarises analytics from de-identified figures only, and names the figures each statement rests on. Severity and other evidence decisions are made by fixed rules, not by an AI model, because evidence must be reproducible.

References

Electronic Communications and Transactions Act 25 of 2002 (South Africa).

Hedera. (n.d.). *Consensus service*. Hedera documentation. https://docs.hedera.com/guides/docs/hedera-api/consensus-service

National Institute of Standards and Technology. (2015). *Secure hash standard (SHS)* (FIPS PUB 180-4). https://doi.org/10.6028/NIST.FIPS.180-4

Protection of Personal Information Act 4 of 2013 (South Africa).

Schwaber, K., & Sutherland, J. (2020). *The Scrum guide*. https://scrumguides.org/scrum-guide.html

South African Police Service. (2025). *Annual report 2024/2025*. https://www.saps.gov.za/about/stratframework/annual_report/2024_2025/2024-25-SAPS-Annual-Report.pdf

Team05. (2026a). *FreightProof SA: Iteration 2 documentation* [Unpublished project document]. INF4027W, University of Cape Town.

Team05. (2026b). *FreightProof SA: Iteration 3 documentation* [Unpublished project document]. INF4027W, University of Cape Town.

Transported Asset Protection Association. (2023). *Trucking security requirements (TSR) 2023*.

Appendix A: Sample Test Cases

Each case below is an automated test in the repository and runs in CI on every pull request. The evidence column names the test file and function. Results are from the full test run on 19 October 2026 shown in Figure 24.

Table 36: Sample test cases

ID	Requirement	Steps and test data	Expected result	Evidence	Result
TC-01	SO-3, TLM 1.2	Create a trip from a mock manifest; then request an unknown manifest number	Trip saved with its manifest key and verifies against its journey lock; unknown manifest returns 404	test_trip_from_pp_manifest.py: test_create_persists_a_manifest_trip; test_unknown_manifest_is_404	Pass
TC-02	SO-3, TLM 1.2 alt. 1c	Two dispatchers create a trip from the same manifest at the same instant (barrier)	Exactly one trip is created; the other request is refused cleanly	test_creation_concurrency.py: test_concurrent_creation_from_one_manifest_creates_one_trip	Pass
TC-03	SO-3, TLM 1.13	Search a barcode belonging to another organisation, and an unknown barcode	Both return the same "not found" response	test_parcel_trace.py: test_foreign_and_unknown_records_are_indistinguishable	Pass
TC-04	SO-1, BR-2	Raise a critical exception, cancel the trip, then review the exception	Each action has its own anchored event and receipt; earlier receipts unchanged	test_fp290_incident_chain.py: test_each_action_has_its_own_event_and_receipts_stay_separate	Pass
TC-05	SO-1, R-07	Repeat TC-04 with Hedera unavailable	Events stored with anchor status "failed"; the evidence itself is intact	test_fp290_incident_chain.py: test_hedera_outage_leaves_failed_events_and_the_source_intact	Pass
TC-06	SO-6, CC 1.11	Arrival with a seal number different from the departure seal	Exactly one critical seal exception is raised	test_arrival_phase.py: test_arrival_seal_number_mismatch_raises_one_critical_exception	Pass
TC-07	SO-5, BR-2	Dispatcher B claims an exception already claimed by A; then A and B submit reviews simultaneously	Claim refused with 409; exactly one review recorded	test_exception_claims.py: test_claim_over_a_colleague_is_409; test_exception_review_concurrency.py: test_simultaneous_reviews_record_exactly_one_dispatcher	Pass
TC-08	SO-4, R-13	Two phones redeem the same QR link at the same instant	Exactly one confirmation succeeds	test_handover_token_concurrency.py: test_two_simultaneous_redemptions_of_the_same_token_produce_exactly_one_success	Pass
TC-09	SO-4, R-13	A second browser opens a link that the receiver's browser holds	The second browser is refused; the receiver's link still works	test_handover_endpoints.py: test_a_refused_binding_does_not_burn_the_real_receivers_token	Pass
TC-10	SO-4, BR-7	Receiver declines the identity check or has no document	Delivery is still confirmed; outcome recorded as unverified	test_handover_endpoints.py: test_full_walk_receiver_without_an_id_still_confirms	Pass
TC-11	R-14	Apply the database lockdown migration and query as the public API roles	Row-level security on; direct access revoked	test_data_api_lockdown.py: test_upgrade_enables_rls_and_revokes_api_roles	Pass
TC-12	R-05	Queue a phase offline and retry it after a lost response	The same idempotency key is sent on every attempt; the server records it once	useOfflineQueue.test.ts: "sends the same idempotency_key on a retry as on the first attempt"	Pass

ID	Requirement	Steps and test data	Expected result	Evidence	Result
TC-13	SO-8, BR-9	Open a revoked audit-pack link	410 response; the attempt is logged	test_audit_pack_public.py: test_open_shared_pack_revoked_is_410_and_logged	Pass
TC-14	SO-8, BR-9	Open an e-mailed audit pack without entering the code	401 response; the attempt is logged	test_audit_pack_email_codes.py: test_open_emailed_pack_without_grant_is_401_and_logged	Pass
TC-15	BP-3, TLM 1.2	Request availability for a horse already on an overlapping trip	The horse is listed as busy, naming the trip; other organisations' trips never appear	test_trip_availability_endpoint.py: test_availability_lists_the_busy_driver_horse_and_trailer	Pass

Appendix B: Addendum – Changes Since Iteration 3

Table 37 lists the main changes to this document since Iteration 3 and the feedback that prompted each. Codes A1 to A32 refer to the comments on the Iteration 3 document; "Marksheet" refers to the Iteration 3 marksheet.

Table 37: Changes since Iteration 3 and the feedback they answer

Feedback	Comment on Iteration 3	Change in Iteration 4 (section)
A1, A2	Acronyms not written out first; organisation citation not in APA style	Acronyms written out on first use; SAPS cited as an organisation in APA 7 style (§1.1); five sources added (References)
A3	No short summary of the business problems	Summary table opens the section (Table 3, §2)
A4	Problems traced to objectives before the objectives were defined	Traceability moved after the objectives (§3.4)
A5; Marksheet	Sponsorship unclear; business objectives framed for LFG	One sponsorship statement (§1.2); business objectives rewritten for FreightProof as a venture, with measures (§3.1)
A6, A17	Parcel detail too high level: manifests, intermediate stops, damage, pallets	Manifest → waybill → parcel hierarchy; pickup and delivery stops per waybill; damage recorded against consignment and vehicles; pallets stated as out of scope (§6.2)
A7	Override needs business rules, approvals, documents, insurer notification	Business rules BR-3 and BR-4 (§6.4)
A8, A13	Dead, forgotten or stolen phone; connectivity and outage testing	BR-6 (§6.4); alternative flows (§6.5); failure-scenario testing (§9.3)
A9	Who is informed; what the customer must do; a receiver who refuses	BR-3 and BR-5 (§6.4); CC 1.8 alt. 5a (§6.5)
A10	"Organisation" and the system boundary unclear	Organisation defined (§1.3, Glossary); boundary and outside actors (§6.3)
A11, A12	Receiver verification and access unclear; authorised receivers	BR-7 (§6.4); CC 1.10 replacement steps (§6.5); identity check opt-in (FP-293)
A14	State machine: refused activation and missing false branch; current stop could not load	Guards and branches redrawn; Arrival added (§6.6)
A15	Architecture looked like a package diagram	Deployment view with nodes and protocols (§7.1)
A16	"Fingerprinted" confused with "digitally signed"	Terminology defined (§7.2)
A18, A22	20 s QR rotation and 200 m geofence not explained	Rationale for every configured value (§7.3); BR-11
A19	Risk status unclear; register split over several pages	One landscape table with a status legend (Table 9, §4)
A20	Driver expected to report offline-queue failures	Automatic dispatcher alert (R-05, §4)
A21	Some risks were features; project risks missing	Feature gaps moved to known limitations (§5.4); project risks R-19 to R-23 (§4)
A23	Trip timeline layout; summary view for many trucks	Live trip list (§8.2); redesigned timeline (§8.3)
A24	Exception handling beyond typing	Claim and review workflow, rules and incident declarations (§6.4, §8.4)
A25	Real identity document in screenshots	Screenshots withdrawn (§8.6)
A26	Analytics not designed for daily use; incident map; patterns	Today and Trends views; sources stated; AI brief (§8.5)
A27	Cleared app data or cache	Alternative flow 4b in CC 1.11 (§6.5); Table 28 (§9.3)
A28	Legal standing of a signature image	ECTA analysis (§7.4)
A29	Empty header in the risk-to-test matrix	Matrix rebuilt with risk IDs (§9.4)
A30	Backlog story hard to follow (FP-144)	Backlog items written as user stories (§10.2, §10.3)
A31	Sprint lengths differ	Velocity normalised per sprint day (§10.4)
A32	Remaining work rose on the burndown	Burndown drawn from Jira with a fixed scope line and explained (§10.2)

Feedback	Comment on Iteration 3	Change in Iteration 4 (section)
Marksheet: testing	Add evidence of automated testing and results	CI pipeline and test counts (§9.2); test cases with results (Appendix A)
Marksheet: professional	One typing standard; empty headers; typos	UK English throughout; no empty headings; captions and cross-references regenerated

Appendix C: Portfolio of Evidence and User Log

Each member's contribution over the project is summarised below, followed by the contribution scores agreed by the team. Each log is reviewed and signed off by the other three members.

TEAM TO DO: The descriptions below were drafted from Jira assignments and design-note authorship. Each member must check and expand their own description (meeting minutes, research, presentation and testing work), agree the scores, and sign the table at the end.

Ciaran Formby (Product Owner). *Code:* manifest-first trip creation (FP-281), the exception inbox with claims and batch review (FP-280), the Arrival phase with every-phase anchoring and trailer geofences (FP-270), anchored critical exceptions, reviews and cancellations (FP-290), the live trip list and live phase timelines (FP-287, FP-288, FP-289), the database lockdown (FP-271), driver substitution (FP-83) and the demo panel (FP-272). *Documentation:* design notes for the Arrival phase and the exception workflow; owner of the final submission package (FP-303). *Other:* team contact for the Pulsit and Parcel Perfect partners; backlog ordering and sprint goals.

Tim Gultig (Scrum Master). *Code:* insurer audit packs, the client portal and incident declarations (FP-279, FP-291, FP-292), receiver confirmation enforcement (FP-273), offline evidence retention (FP-276), seal reading from photographs (FP-300), precinct address search (FP-309), selfie storage (FP-316), security hardening (FP-308) and the identity-check layering fix (FP-277). *Documentation:* the insurer audit-trail design, including research on claim forms, TAPA TSR and ECTA. *Other:* sprint ceremonies, Jira reconciliation and the sprint records.

Thomas Davis. *Code:* cost analytics (FP-294), today's incident map and figure sources (FP-296), the AI insights brief (FP-295), and the vehicle-clash guard and availability strip (FP-139, FP-142). *Documentation:* analytics specifications for cost and provenance; project context and plan sections. *Other:* architecture and package diagrams.

Chiko Kasongo. *Code:* parcel search by barcode (FP-149), per-user rate limits (FP-274), database timeouts (FP-275), the load test (FP-299) and the opt-in receiver identity check (FP-293). *Documentation:* UI evidence and state-machine sections. *Other:* sprint metrics, burndown and velocity reporting.

Table 38: Contribution scores out of 5

Member	Coding	Documentation	Presentation	Research	Meeting minutes	Communication
Ciaran Formby	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Tim Gultig	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Thomas Davis	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
Chiko Kasongo	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

For reference, the Iteration 3 scores were: Ciaran 5, 5, 4, 5, 4, 5; Tim 5, 5, 5, 4, 4, 5; Thomas 5, 5, 4, 4, 4, 5; Chiko 4, 5, 4, 4, 5, 5 (same column order).

Table 39: Sign-off of the user logs

Member's log	Reviewed and signed by	Reviewed and signed by	Reviewed and signed by
Ciaran Formby	Tim Gultig **[sign, date]**	Thomas Davis **[sign, date]**	Chiko Kasongo **[sign, date]**

Member's log	Reviewed and signed by	Reviewed and signed by	Reviewed and signed by
Tim Gultig	Ciaran Formby **[sign, date]**	Thomas Davis **[sign, date]**	Chiko Kasongo **[sign, date]**
Thomas Davis	Ciaran Formby **[sign, date]**	Tim Gultig **[sign, date]**	Chiko Kasongo **[sign, date]**
Chiko Kasongo	Ciaran Formby **[sign, date]**	Tim Gultig **[sign, date]**	Thomas Davis **[sign, date]**